EU Declaration of Conformity

EU Declaration of Conformity

According to

EMC directive 2004/108/EC

For the following

Product:	**RTEX Based Network Controller**
Model Name:	**PCI-Rxx04, RTEX-PM,**
	RTEX-DI32, RTEX-DO32T

Manufactured at :	**AJINEXTEK CO., LTD.**
Address:	**9-3 Horim-dong, Dalseo-gu,**
	Daegu 704-240, Korea

We hereby declare, Electromagnetic Compatibility Directives (2004/108/EC, 92/31/EEC) are fulfilled, as laid out in the guideline set down by the member states of the EEC Commission. This declaration is valid for all samples that are part of this declaration, which are manufactured according to the production charts appendix.

The standards relevant for the evaluation of EMC requirements are as follows:

Test Standards:	**EN 55022:2006+A1:2007, Class A**
	EN 55024:1998+A1:2001+A2:2003

Date of issue: August 08, 2011

AJINEXTEK CO., LTD.

9-3 Horim-dong, Dalseo-gu,
Daegu 704-240, Korea

(Name and signature of authorized person)

EMC TEST REPORT

Test report No:	**EMC-CE-2666**
Type of Equipment:	**RTEX Based Network Controller**
Model Name:	**PCI-Rxx04, RTEX-PM, RTEX-DI32, RTEX-DO32T**
Applicant:	**AJINEXTEK CO., LTD.** **9-3 Horim-dong, Dalseo-gu,** **Daegu 704-240, Korea**
Manufacturer:	**AJINEXTEK CO., LTD.** **9-3 Horim-dong, Dalseo-gu,** **Daegu 704-240, Korea**
Test standards:	**EN 55022:2006+A1:2007, Class A** **EN 55024:1998+A1:2001+A2:2003**
Testing Laboratory:	**EMC Compliance Ltd.**
Test result:	**Complied**

This product complies with the requirements of the EMC Directive 2004/108/ EC.
The results in this report apply only to the sample tested.
This test report shall not be reproduced, except in full, without the written approval of EMC compliance Laboratory.

Date of receipt: 2011. 07. 12

Date of testing: 2011. 07. 21 ~ 07. 26

Issued date: 2011. 08. 08

Tested by: KIM, IN-HO

Approved by: YEOM, HAN-SEOK

This test report shall not be reproduced, except in full, without the written approval.

Contents

1. Applicant information 3

2. Laboratory information 4

3. Test system configuration 5

3.1 Operation environment 5

3.2 Measurement Uncertainty 6

4. Description of E.U.T. 7

4.1 General information 7

4.3 Auxiliary equipments 9

4.4 Test configuration 10

4.5 Operating conditions 11

5. Summary of test results 12

5.1 Modification to the E.U.T. 12

5.2 Summary of EMI emission test results 12

5.3 Summary of immunity test results 12

5.4 Performance criteria 13

6. Test results 14

6.1 Conducted Emission 14

6.2 Radiated Emission 20

6.3 Electrostatic Discharge 24

6.4 Radio Frequency Electromagnetic Fields 29

6.5 Electric Fast Transient/BURST 31

6.6 Surge 34

6.7 Conducted Immunity 36

6.8 Dips and Interruptions 39

7. E.U.T. photographs 41

This test report shall not be reproduced, except in full, without the written approval.

1. Applicant information

Applicant:	AJINEXTEK CO., LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone:	+82-53-593-3700
Fax:	+82-53-593-3703
E-mail:	doriwon@ajinextek.com
Contact name:	**Son yong hoon**

Manufacturer:	AJINEXTEK CO., LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone:	+82-53-593-3700
Fax:	+82-53-593-3703
E-mail:	doriwon@ajinextek.com
Contact name:	**Son yong hoon**

This test report shall not be reproduced, except in full, without the written approval.

2. Laboratory information

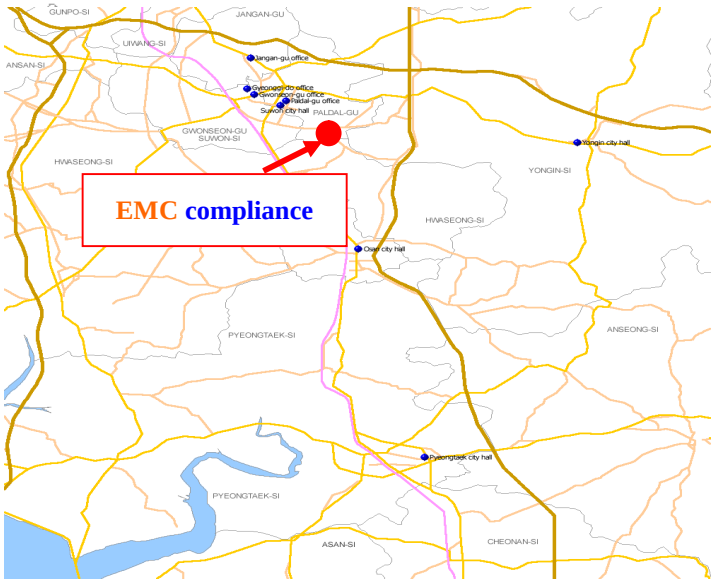
Address

EMC compliance Ltd.
480-5 Sin-dong, Yeongtong-gu, Suwon-city, Gyeonggi-do, 443-390, Korea
Telephone Number: 82 31 336 9919
Facsimile Number: 82 31 336 4767

FCC CAB.: KR0040
VCCI Registration No. : R-3327, G-198, C-3706, T-1849
Industry Canada Registration No. : 8035A
KOLAS NO.: 231

SITE MAP





This test report shall not be reproduced, except in full, without the written approval.

3. Test system configuration

3.1 Operation environment

		Temperature	Humidity	Pressure
Chamber(10 m)	:	27 °C	30 % R.H.	-
Shielded room(CE)	:	27 °C	30 % R.H.	-
Shielded room(ESD)	:	26 °C	45 % R.H.	100.3 kPa

Test site

These testing items were performed following locations;

Shielded Room : Conducted Emission
ESD, EFT/Burst, Surge,
CS, Dip/Interruption
Chamber (10 m) : Radiated Emission
Fully anechoic chamber (3 m) : RS

This test report shall not be reproduced, except in full, without the written approval.

3.2 Measurement Uncertainty

All measurements involve certain levels of uncertainties, especially in field of EMC. The factors contributing to uncertainties are test receiver, cable loss, antenna factor calibration, Antenna directivity, antenna factor variation with height, antenna phase center variation, antenna frequency interpolation, measurement distance variation, site imperfection, mismatch, and system repeatability. Based on CISPR 16-4-2, the measurement uncertainty level with a 95 % confidence level was applied.

Conducted emission measurement :(k = 2, 95 %)

9 ㎑ ~ 150 ㎑: ± 3.77 [㏈]

150 ㎑ ~ 30 ㎒: ± 3.35 [㏈]

Radiated Emission measurement :(k = 2, 95 %)

30 ㎒ ~ 300 ㎒: 3 m: ± 4.3 [㏈]

10 m: ± 4.3 [㏈]

300 ㎒ ~ 1 000 ㎒: 3 m: ± 4.4 [㏈]

10 m: ± 4.3 [㏈]

Radio Frequency Electromagnetic Fields :(k = 2, 95 %)

± 1.82 [㏈]

This test report shall not be reproduced, except in full, without the written approval.

4. Description of E.U.T.

4.1 General information

* PCI-Rxx04

Items	Specification
Communication type	Real Time Express(Panasonic distributed network specification)
Communication port	RJ45 TX-RX 1EA
Number of control axis	16axis BASE, 32axis (Option Contact Board)
Motion control chip set	CAMC-QI 4EA (Expend CAMC-QI 4EA)
Output frequency	~10.0Mpps (Standard of Clock 39.3216MHz)
LED display	Power(Yellow), Link(Green), Error(Red), Com(Green)
Number of configuration slave	Max 32 RTEX Slave Module
Data Bus Width	32bit data mode
Bus Interface	PCI 2.2, Memory Access
Internal current consumption	MAX. 1.5A (5Vdc)
Input/output for digital control	IN2, IN3 : Photo-coupler(24VDC Level)
	OUT2, OUT3 : Photo-coupler + Drive IC(24VDC Level)
Operating temperature / Humidity	Temperature 0 ~ 55°C / Humidity 5 ~ 90% (no condensation)
Storage temperature	-20 ~ 70°C
Weight	120g (Expend 32axis-161g)
Exterior dimension of product	174.63 (W) mm x 106.00 (L) mm

* RTEX-DI32

Items	Specification
Module function	RTEX based 32axes of slave digital input function module
Module ID setting	Decimal number rotary switch x 2 (0 ~ 31)
Mounting type	Standard 35mm DIN rail mounting
Internal current consumption	Max 500mA, 24Vdc (connected to TB1 outside the module)
LED display	Power(Yellow), Live(Green), Error(Red), Com(Yellow)
	IN1 ~ IN32(Yellow)
Input for digital control	Input 32 channels, 24VDC Level
	IN1~IN32 : Photo-coupler isolation (More over MIN. 2mA/channel)
	PCOM, NCOM : Digital I/O Positive and Negative Common
	Connector : Phoenix 16 X 2 (MCD 1,5/16-GIF-3,81)
Operating temperature / Humidity	Temperature 0 ~ 55°C / Humidity 5 ~ 90% (no condensation)
Storage temperature	-20 ~ 70°C
Weight	245g
Exterior dimension of product	100 (H) mm X 90 (D) mm X 40 (W) mm

This test report shall not be reproduced, except in full, without the written approval.

* RTEX-DO32T

Items	Specification
Module function	RTEX based 32axes of slave digital output function module
Module ID setting	Decimal number rotary switch x 2 (0 ~ 31)
Mounting type	Standard 35mm DIN rail mounting
Internal current consumption	Max 500mA, 24Vdc (connected to TB1 outside the module)
LED display	Power(Yellow), Live(Green), Error(Red), Com(Yellow)
	OUT1~OUT32(Red)
Input for digital control	Output 32 channels, 24VDC Level
	IN1~IN32 : Photo-coupler isolation (Below MAX. 50mA/channel)
	PCOM, NCOM : Digital I/O Positive and Negative Common
	Connector : Phoenix 16 X 2 (MCD 1,5/16-GIF-3,81)
Operating temperature / Humidity	Temperature 0 ~ 55°C / Humidity 5 ~ 90% (no condensation)
Storage temperature	-20 ~ 70°C
Weight	245g
Exterior dimension of product	100 (H) mm X 90 (D) mm X 40 (W) mm

* RTEX-PM

Items	Specification
Module function	RTEX based 1axes of slave motion function module
Module ID setting	Decimal number rotary switch x 2 (0 ~ 31)
Mounting type	Standard 35mm DIN rail mounting
Internal current consumption	Max 500mA, 24Vdc (connected to TB1 outside the module)
LED Disply	Power(Yellow), Live(Green), Error(Red), Com(Yellow)
Pulse output	PULSE+, PULSE-, DIR+, DIR
	LINE TRANSMITTER (5VDC, Max 10MPPS)
Encoder input	ENC-A, /A, B, /B, Z, /Z : High-speed Photo-coupler (5VDC, Max 8MPPS)
	ENCPWR, DGND : 5VDC output (when using the encoder for step motors)
Input/output for motion control	ALARM, INP, RDY(IN4) : Photo-coupler (24VDC Level)
	SVON, ALMC, DCC, TRG(*) : Dalington Photo-coupler (24VDC Level)
	LIMIT+, LIMIT-, ORG : Photo-coupler (24VDC Level)
	TRG+, TRG- : Line Transmitter(5VDC Level)
	BRK(OUT4) : Photo-coupler + Drive IC (24VDC Level)
	PCOM, NCOM : Digital I/O Positive and Negative Common
	Motion Connector : Honda 26pin X 1 EA
	Motion I/O Connector : 3M 36pin MDR (N10236-52A2PL) X 1 EA
Universal input/output	IN2, IN3 : Photo-coupler(24VDC Level)
	OUT2, OUT3 : Photo-coupler + Drive IC (24VDC Level)
Input of Emergency stop	Emergency stop of outside : Photo-coupler isolation (24VDC Level)
	Connector : Pheonix Contact 5 pin, 3.5mm pitch
Operating temperature / Humidity	Temperature 0 ~ 55°C / Humidity 5 ~ 90% (no condensation)
Storage temperature	-20 ~ 70°C
Weight	192g
Exterior dimension of product	120mm (H) x 110 (D) mm x 25 (W) mm

This test report shall not be reproduced, except in full, without the written approval.

4.2 Product description

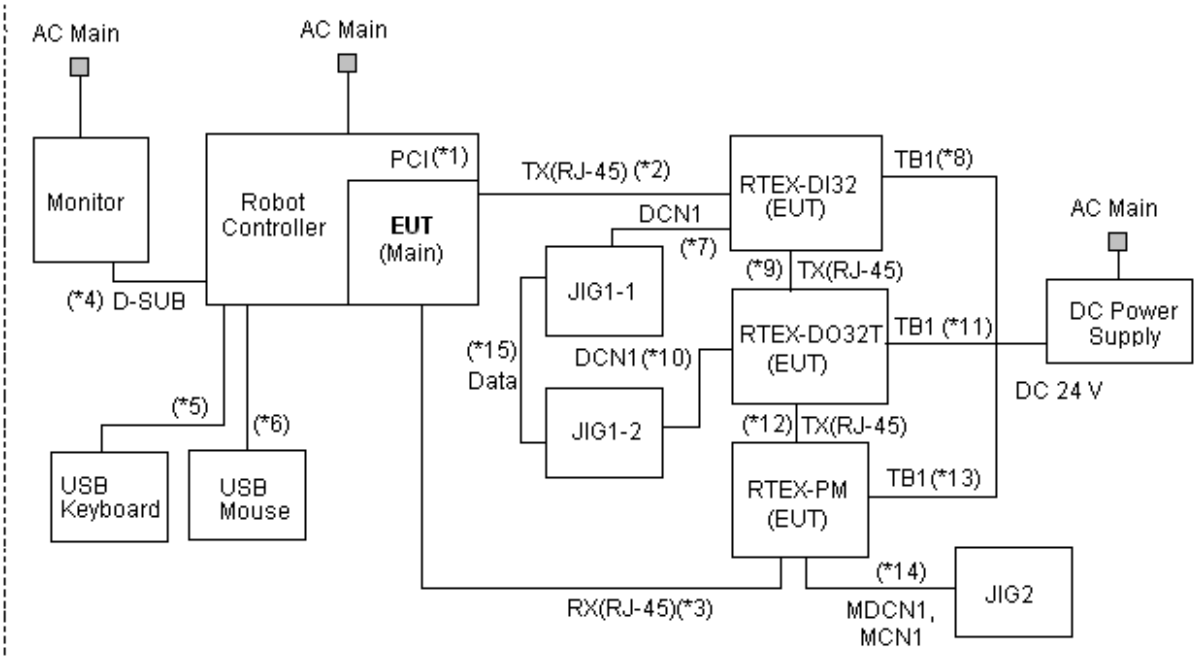
Type of product	RTEX Based Network Controller
Model name (Basic)	PCI-Rxx04, RTEX-PM, RTEX-DI32, RTEX-DO32T
Model name (Variant)	N/A
Difference	-
Trade name	-
Serial no	Engineering Sample
Testing voltage	AC main(230 V), DC 24 V
Product rating	AC main(230 V), DC 24 V
Note	* PCI-Rxx04: AC main(230 V) RTEX-PM, RTEX-DI32, RTEX-DO32T: DC 24 V

4.3 Auxiliary equipments

Type	Model / Part #	Serial number	Manufacturer
Monitor	CX501N	N500HVFY604277V	SAMSUNG
USB Keyboard	SDM4600UH	4S011446	SAMSUNG
USB Mouse	M-UV69a	HCA51201503	LG
Robot Controller	ARC-II	-	AJINEXTEK CO., LTD.
DC Power Supply	-	-	-
JIG1(2EA)	-	-	-
JIG2	-	-	-

This test report shall not be reproduced, except in full, without the written approval.

4.4 Test configuration



Note	Start		End		Cable	
*	Name	I/O port	Name	I/O port	Length (m)	Spec.
1	**EUT** (Main)	PCI	ARC-II	PCI	Direct	-
2		TX(RJ-45)	RTEX-DI32 (EUT)	RX(RJ-45)	3.0	Shield
3		RX(RJ-45)	RTEX-PM (EUT)	TX(RJ-45)	3.0	Shield
4	Robot Controller	D-SUB	Monitor	D-SUB	1.8	Shield
5		USB	USB Keyboard	USB	1.8	Shield
6		USB	USB Mouse	USB	1.8	Shield
7	RTEX-DI32 (EUT)	DCN1	JIG1-1	DCN1	0.2	Non-Shield
8		TB1	DC Power Supply	TB1	1.5	Non-Shield
9		TX(RJ-45)	RTEX-DO32T (EUT)	RX(RJ-45)	3.0	Shield
10	RTEX-DO32T (EUT)	DCN1	JIG1-2	DCN1	0.2	Non-Shield
11		TB1	DC Power Supply	TB1	1.5	Non-Shield
12		TX(RJ-45)	RTEX-PM (EUT)	RX(RJ-45)	3.0	Shield
13	RTEX-PM (EUT)	TB1	DC Power Supply	TB1	1.5	Non-Shield
14		MDCN1, MCN1	JIG2	MDCN1, MCN1	2.5	Non-Shield
15	JIG1-1	Data	JIG1-2	Data	2.0	Non-Shield

This test report shall not be reproduced, except in full, without the written approval.

4.5 Operating conditions

The EUT was configured as normal intended use.

Test mode	Normal operating
1	EUT operating (Control) test mode. (Program: EZ Config PM)

* Note: 2 types of powers are available for the product, that are AC main(230 Ⅴ), DC 24 Ⅴ.
Therefore, tests were performed for 2 different types of powers.

This test report shall not be reproduced, except in full, without the written approval.

5. Summary of test results

5.1 Modification to the E.U.T.

None

5.2 Summary of EMI emission test results

Application	Test method	Test result
Conducted Emission - AC main port	EN 55022:2006+A1:2007, Class A	Complied
Radiated Emission - AC main, DC 24 Ⅴ	EN 55022:2006+A1:2007, Class A	Complied
Harmonics current	EN 61000-3-2:2006+A2:2009	N/A
Voltage fluctuations and flickers	EN 61000-3-3:2008	N/A

5.3 Summary of immunity test results

Items	Application	Test method	Test result
Electrostatic discharge	Enclosure	EN 61000-4-2:2009	Complied
Radiated RF immunity	Enclosure	EN 61000-4-3:2006	Complied
Electric Fast Transient/BURST	AC main, DC 24 Ⅴ Signal	EN 61000-4-4:2004	Complied
Surge	AC main	EN 61000-4-5:2006	Complied
Conducted RF immunity	AC main, DC 24 Ⅴ Signal	EN 61000-4-6:2007	Complied
Magnetic field immunity	Enclosure	EN 61000-4-8:2010	N/A
Voltage dip/interruption	AC main	EN 61000-4-11:2004	Complied

This test report shall not be reproduced, except in full, without the written approval.

5.4 Performance criteria

Performance criterion A: The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended.
The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B: After the test, the equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test.
If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as untended.

Performance criterion C: Loss of function is allowed, provided the function is self-recoverable or can be restored by the operating of the controls by the user In accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

This test report shall not be reproduced, except in full, without the written approval.

6. Test results

6.1 Conducted Emission

Test specification	EN 55022:2006+A1:2007, Section 5, Class A		
Test mode	EUT operating test mode.		
Date	2011. 07. 21		
Testing Voltage	AC main, DC 24 V		
Test facility	Shielded room (CE#1)		
Temperature (°C)	27 °C	Humidity (% R.H.)	30 % R.H.
Remarks	Complied AC Main Minimum limit margin is 6.76 dB at 0.555 ㎒. (Average)		

6.1.1 Limits of conducted emission measurement

- AC main

Frequency [㎒]	Class A (dB(㎶))		Class B (dB(㎶))	
	Quasi-peak	Average	Quasi-peak	Average
0.15 ~ 0.5	79	66	66 ~ 56 *	56 ~ 46*
0.5 ~ 5	73	60	56	46
5 ~ 30	73	60	60	50

*The limit decreases linearly with the logarithm of frequency.

This test report shall not be reproduced, except in full, without the written approval.

6.1.2 Measurement procedure

The measurements were performed in a shielded room. EUT was setup as shown in photograph and placed on a non-metallic table height of 0.8 m above the reference ground plane. The rear of table was located 0.4 m to the vertical conducted plane. EUT was power through the LISN, which was bonded to the ground plane. The LISN power was filtered. Each EUT power lead, except ground (safety) lead was individually connected through a LISN to input power source. EUT signal cables that hung closer than 0.4 m to the Horizontal metal ground 0.3 m ~ 0.4 m long. The power cord was bundles in the center. All peripheral equipment was powered from a sub LISN. The LISN and ISN were positioned 0.8 m from the EUT. Peak and Average detection were used in preliminary testing and Quasi-peak and Average detections were used at final measurement.

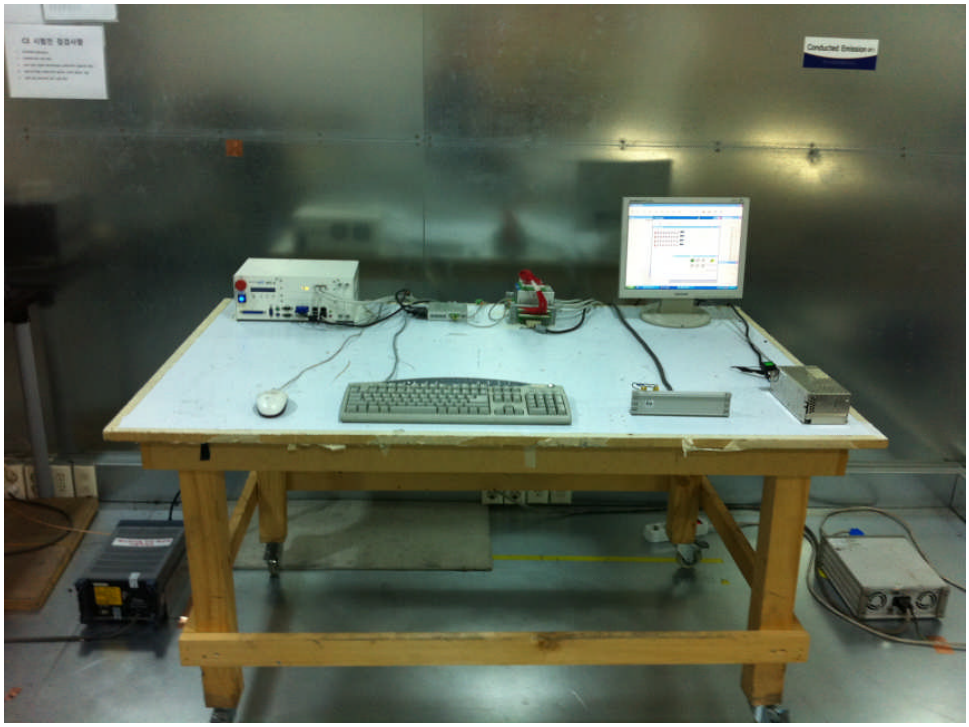
6.1.3 Used equipments

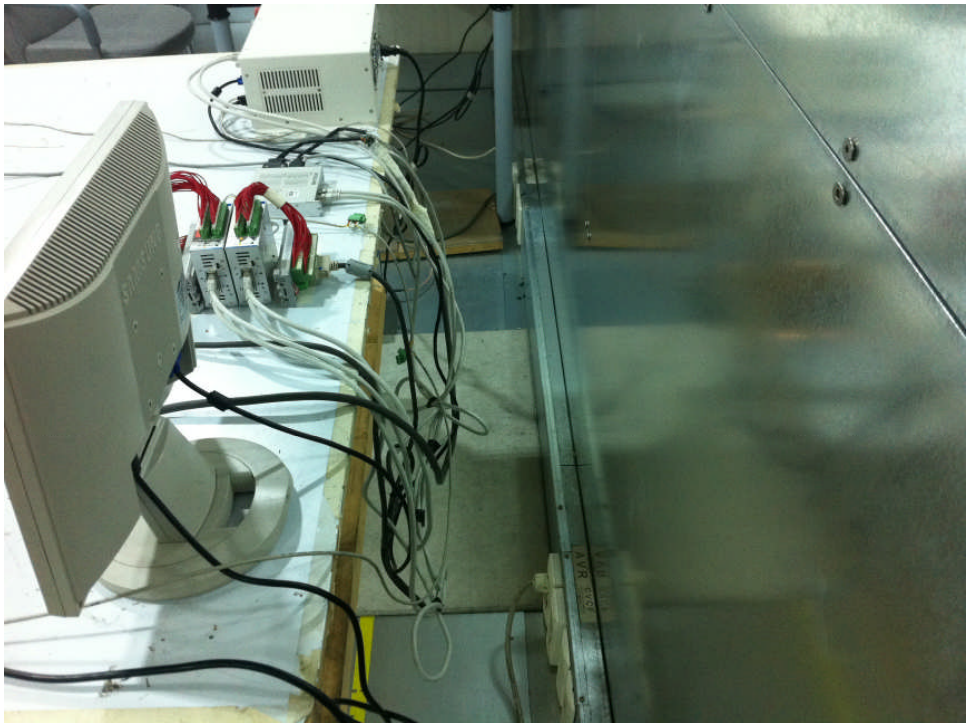
Equipment	Model	Serial No.	Makers	Next Cal. Date	Used
Test Receiver	ESHS10	843276/003	R&S	12.06.14	☒
LISN	ESH3-Z5	100267	R&S	12.07.08	☒
LISN	L2-16A	0000J10705	PMM	-	☒

This test report shall not be reproduced, except in full, without the written approval.

6.1.4 Photographs of test setup

* AC main





This test report shall not be reproduced, except in full, without the written approval.

6.1.5 Conducted emission measurement result

* AC main(230 V)

Frequency	Correction Factor		Line	Quasi-peak			Average		
				Limit	Reading	Result	Limit	Reading	Result
[㎒]	LISN	Cable		[dB(㎶)]	[dB(㎶)]	[dB(㎶)]	[dB(㎶)]	[dB(㎶)]	[dB(㎶)]
0.207	0.07	0.01	N	79.00	61.31	61.39	66.00	55.49	55.57
0.276	0.08	0.01	H		55.63	55.72		52.27	52.36
0.417	0.07	0.01	N		46.90	46.98		43.36	43.44
0.486	0.08	0.01	H		53.45	53.54		51.29	51.38
0.555	0.08	0.01	H	73.00	55.53	55.62	60.00	53.15	53.24
0.624	0.08	0.01	H		54.63	54.72		52.03	52.12
0.900	0.09	0.03	H		54.22	54.34		51.19	51.31
0.969	0.09	0.03	H		54.64	54.76		51.53	51.65

This test report shall not be reproduced, except in full, without the written approval.

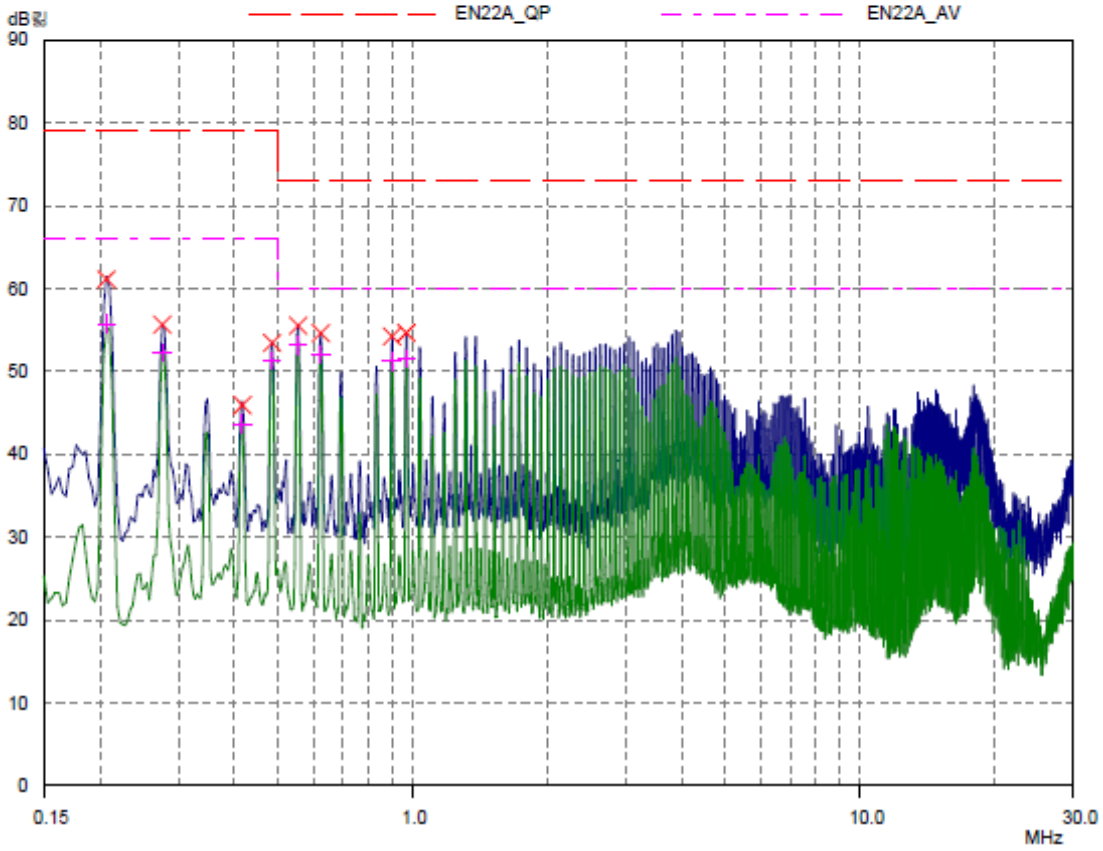
EUT: PCI-Rxx04, RTEX-PM, RTEX-DI32, RTEX-DO32T
Manuf:
Op Cond: H
Operator:
Test Spec: EN55022 Class A Conducted Emission
Comment:

Result File: ajin_h.dat : New Measurement

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB

Final Measurement: Detectors: X QP / + AV
Meas Time: 1sec
Peaks: 8
Acc Margin: 25 dB



This test report shall not be reproduced, except in full, without the written approval.

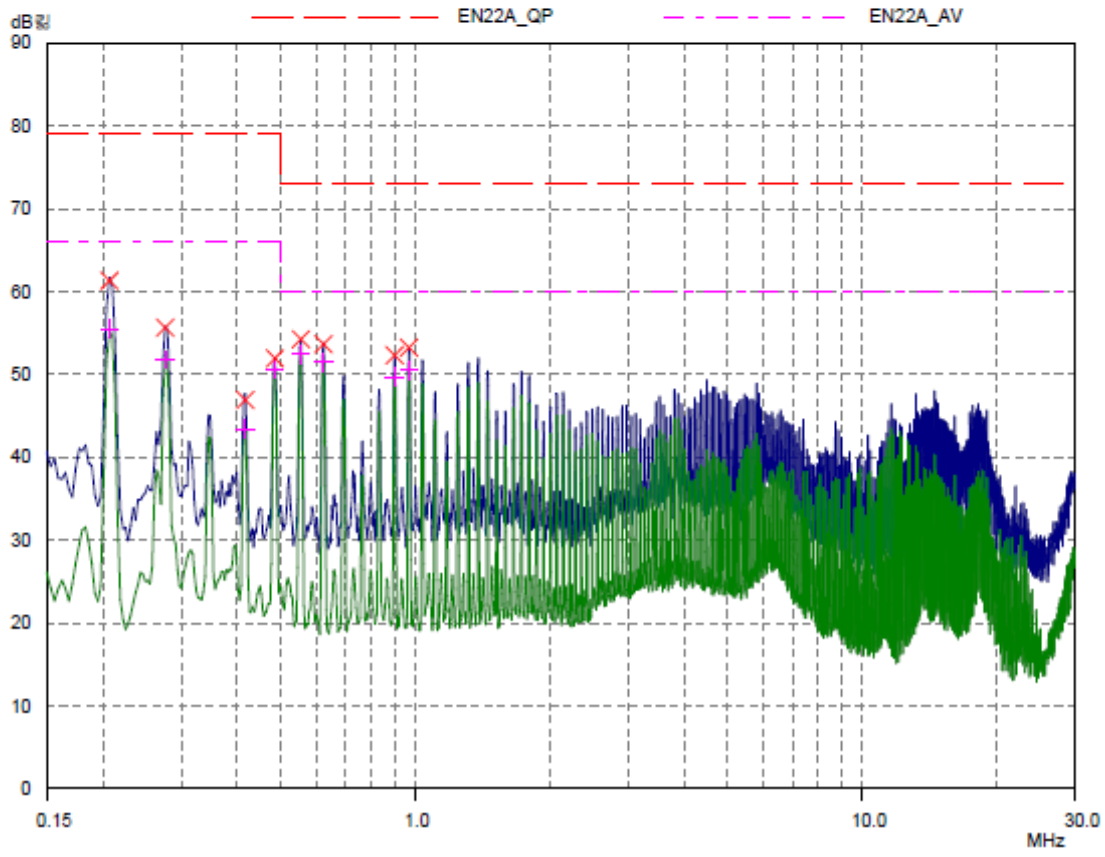
EUT: PCI-Rxx04, RTEX-PM, RTEX-DI32, RTEX-DO32T
Manuf:
Op Cond: N
Operator:
Test Spec: EN55022 Class A Conducted Emission
Comment:

Result File: ajin_n.dat : New Measurement

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	10msec	Auto	OFF	60dB

Final Measurement: Detectors: X QP / + AV
Meas Time: 1sec
Peaks: 8
Acc Margin: 25 dB



This test report shall not be reproduced, except in full, without the written approval.

6.2 Radiated Emission

Test specification	EN 55022:2006+A1:2007, Sections 6, Class A		
Test mode	EUT operating test mode.		
Date	2011. 07. 21		
Testing Voltage	AC main, DC 24 V		
Test facility	10 m Chamber		
Temperature (°C)	27 °C	Humidity (% R.H.)	30 % R.H.
Remarks	Complied Minimum limit margin is 4.5 dB at 375.005 ㎒.		

6.2.1 Limits of radiated emission measurement

Frequency [㎒]	Class A (dB(㎶/m)) @10 m	Class B (dB(㎶/m)) @10 m
30 ~ 230	40	30
230 ~ 1 000	47	37

6.2.2 Measurement procedure

The test was done at a 10 m chamber with a quasi-peak detector. EUT was placed on EUT was placed on a non-metallic table height of 0.8 m above the reference ground plane. Cables were folded back and forth forming a bundle 0.3 m to 0.4 m long and were hanged at a 0.4 m height to the ground plane. Cables connected to EUT were fixed to cause maximum emission. Test was made with the antenna positioned in both the horizontal and vertical planes of polarization. The measurement antenna was varied in height above the conducting ground plane to obtain the maximum signal strength.

This test report shall not be reproduced, except in full, without the written approval.

6.2.3 Used equipments

Equipment	Model no.	Serial no.	Makers	Next cal. date	Used
Test Receiver	ESCI	100710	R&S	11.12.01	☒
Bi-Log Antenna	VULB 9168	9168-440	SCHWARZBECK	13.07.08	☒
Amplifier	310N	293004	SONOMA INSTRUMENT	11.12.01	☒
3 dB Attenuator	8491A	27444	HP	11.11.30	☒
Antenna Mast	MA4000-EP	303	Innco Systems	-	☒
Turn Table	DT2000S-1t	079	Innco Systems	-	☒

6.2.4 Sample calculation

The field strength is calculated adding the antenna Factor, cable loss and, Antenna pad adding, subtracting the amplifier gain from the measured reading.

The sample calculation is as follow:

Result = M.R + C.F(A.F + C.L +3 dB Att – A.G)

M.R = Meter Reading

C.F = Correction Factor

A.F = Antenna Factor

C.L = Cable Loss

A.G= Amplifier Gain

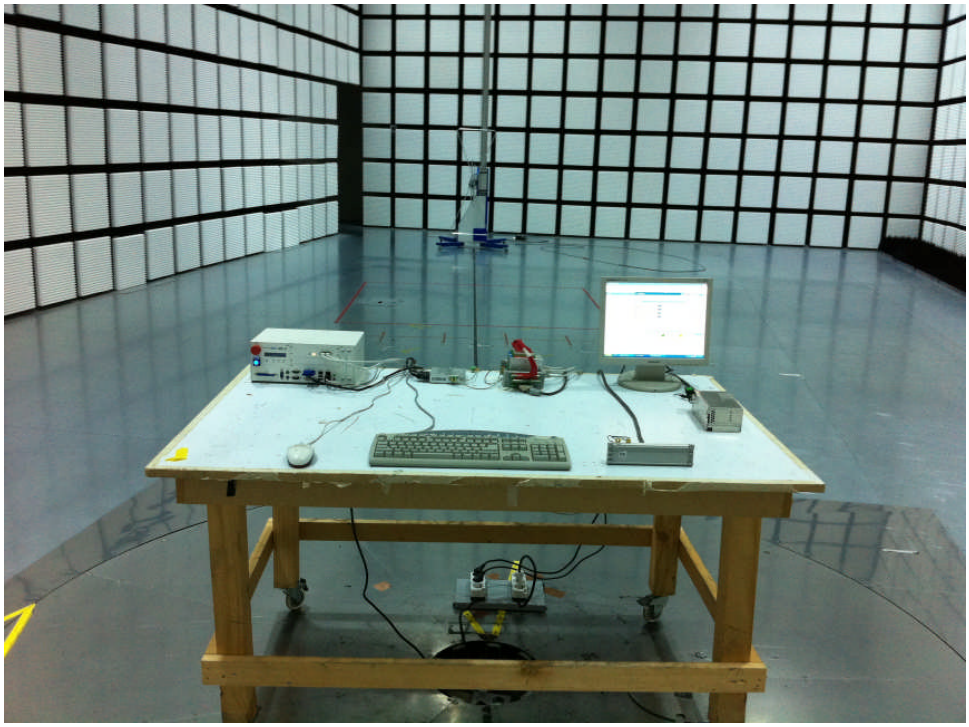
3 dB Att = 3 dB Attenuator

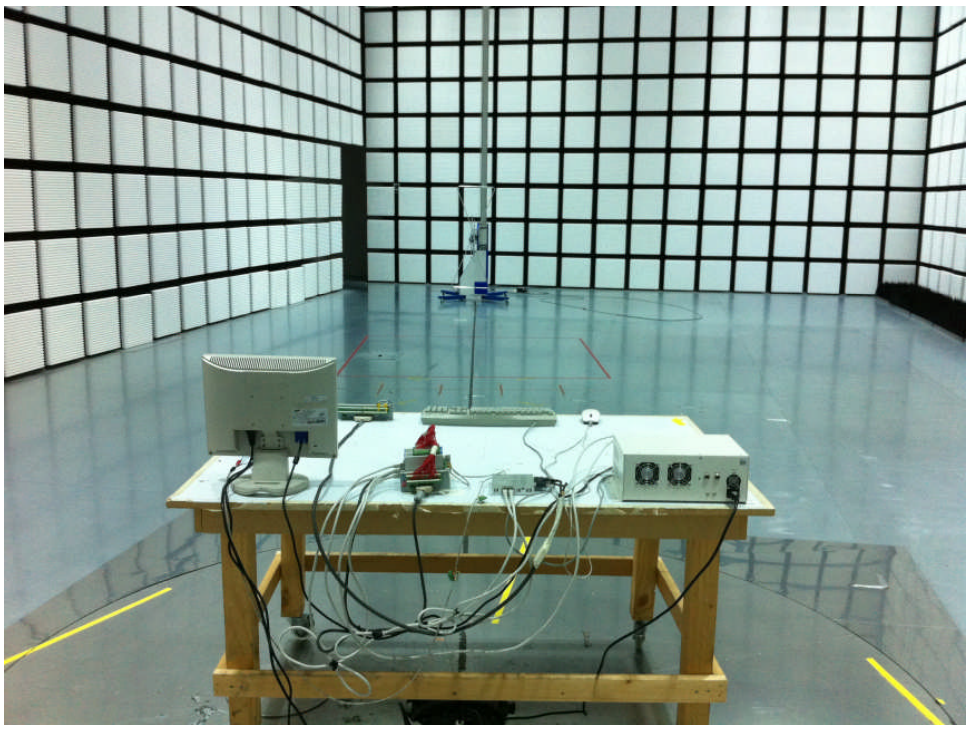
If M.R is 30 dB, A.F 12 dB, C.L 5 dB, 3 dB, A.G 35 dB

The result is 30 + 12 + 5 + 3 – 35 = 15 dB(μV/m)

This test report shall not be reproduced, except in full, without the written approval.

6.2.5 Photographs of test setup

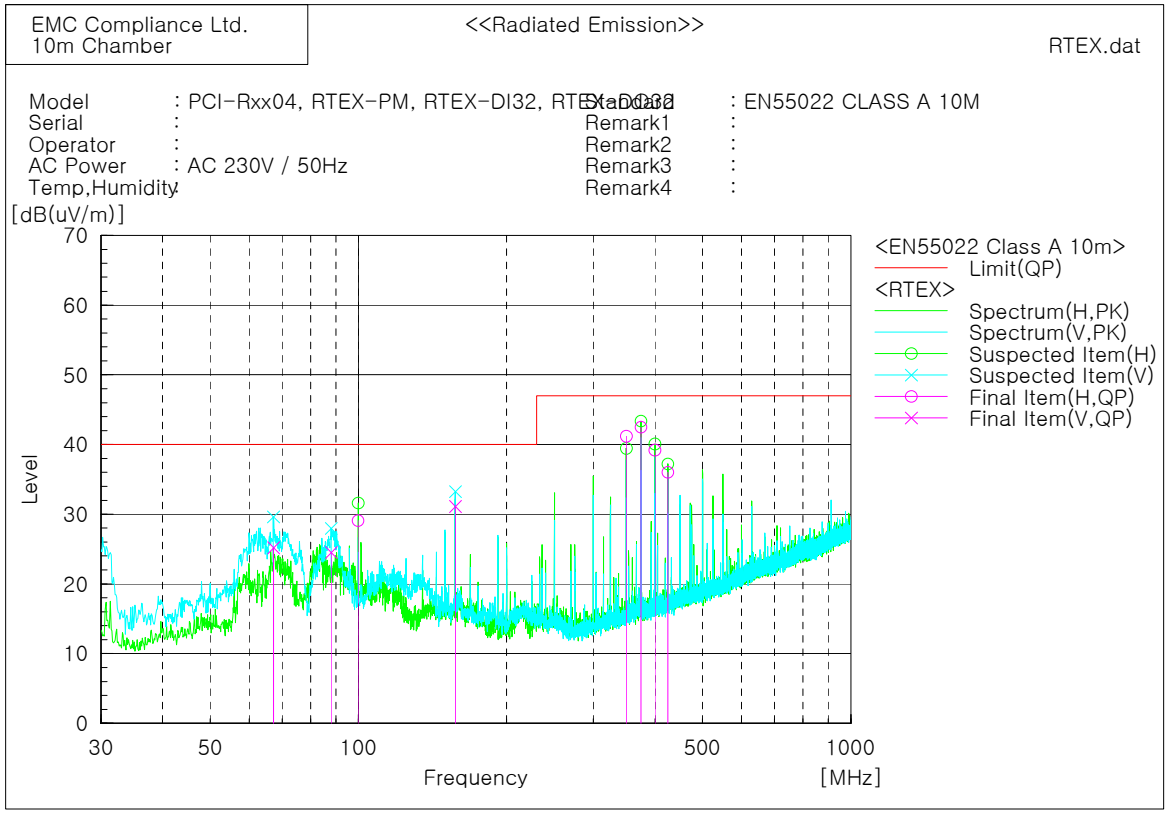




This test report shall not be reproduced, except in full, without the written approval.

6.2.6 Radiated emission measurement result

*** Graph and Data**



Final Result

No.	Frequency [MHz]	(P)	Reading QP [dB(uV)]	c.f [dB(1/m)]	Result QP [dB(uV/m)]	Limit QP [dB(uV/m)]	Margin QP [dB]	Height [cm]	Angle [deg]
1	67.254	V	41.5	-16.3	25.2	40.0	14.8	299.0	18.7
2	88.012	V	42.5	-18.0	24.5	40.0	15.5	100.0	222.2
3	99.890	H	46.2	-17.1	29.1	40.0	10.9	400.0	240.8
4	157.310	V	43.5	-12.4	31.1	40.0	8.9	100.0	196.4
5	350.010	H	51.5	-10.3	41.2	47.0	5.8	302.0	256.7
6	375.005	H	52.1	-9.6	42.5	47.0	4.5	202.0	239.5
7	400.055	H	48.2	-9.0	39.2	47.0	7.8	302.0	269.7
8	425.012	H	44.2	-8.2	36.0	47.0	11.0	202.0	23.0

This test report shall not be reproduced, except in full, without the written approval.

6.3 Electrostatic Discharge

Test specification	EN 61000-4-2:2009, Criteria : B				
Test level	Contact: ± 2 kV, ± 4 kV Air: ± 2 kV, ± 4 kV, ± 8 kV HCP / VCP: : ± 2 kV, ± 4 kV				
Discharge impedance	330 Ω / 150 pF				
Date	2011. 07. 26				
Testing Voltage	AC main, DC 24 V				
Number of discharge (Each polarity)	Contact(HCP / VCP): 25 Air: 10				
Interval between discharges	1 s				
Temperature(°C)	26 °C	Humidity (% R.H.)	45 % R.H.	Pressure (kPa)	100.3 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.3.1 Measurement procedure

A ground reference plane was located on the floor, and connected to earth via a low Impedance connection. The return cable of the ESD generator was connected to the reference plane. In case of floor standing equipment, EUT was placed on the reference plane on 0.1 m of insulating Support. In case of table top equipment, EUT was placed on a wooden table 0.8 m above the reference grounded floor. A horizontal coupling plane (HCP) was placed on the table, and Connected to the reference plane via a 470 kΩ resistor located in each end (0.5 ㎜ insulating support between EUT and HCP). In both cases a vertical coupling plane(VCP) OF 0.5 X 0.5 m was located 0.1 m from the EUT's sides. The VCP was connected to the reference plane in the same matter as the HCP.

6.3.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. Date	Used
ESD Tester	NSG 437	182	TESEQ	12.05.23	☒
HCP	-	-	-	-	☒
VCP	-	-	-	-	☒

This test report shall not be reproduced, except in full, without the written approval.

6.3.3 Photographs of test setup



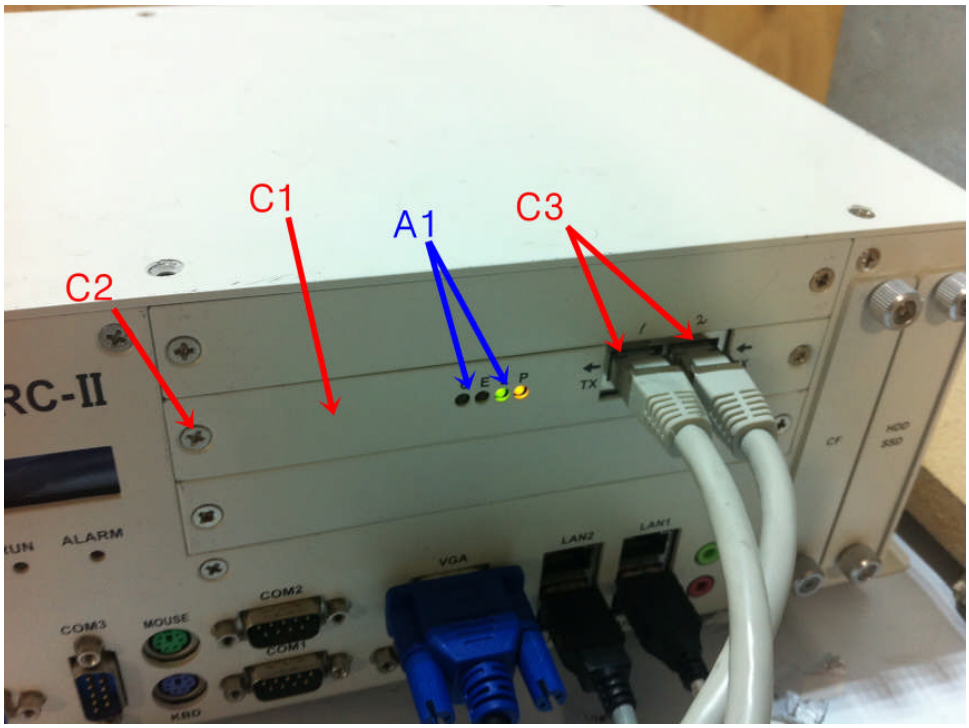
This test report shall not be reproduced, except in full, without the written approval.

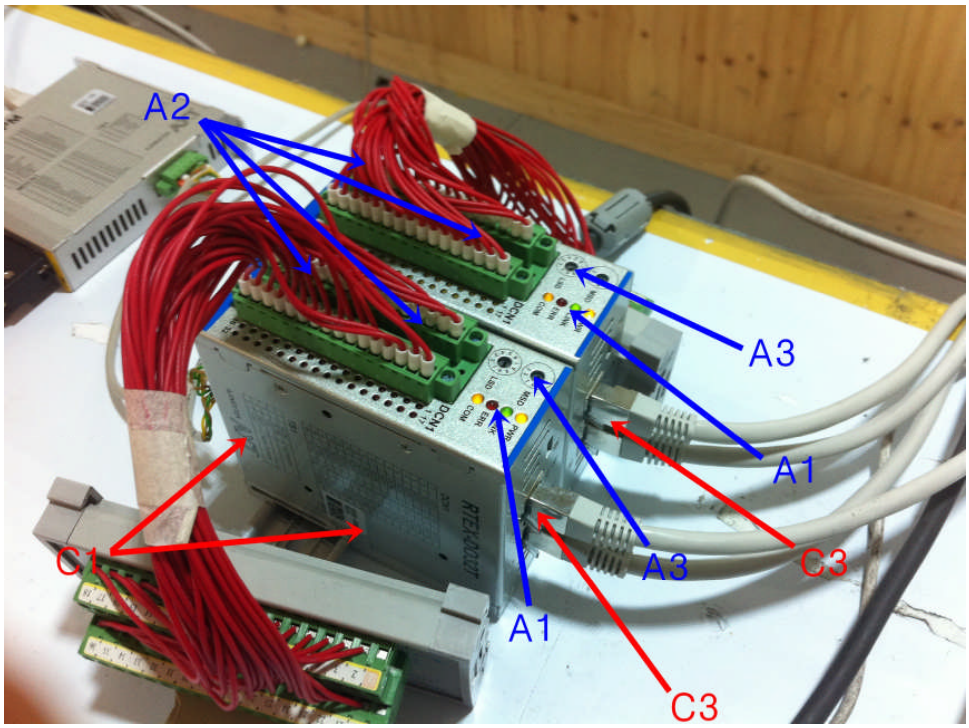
6.3.4 Measurement result

Electrostatic Discharge (Test Point)

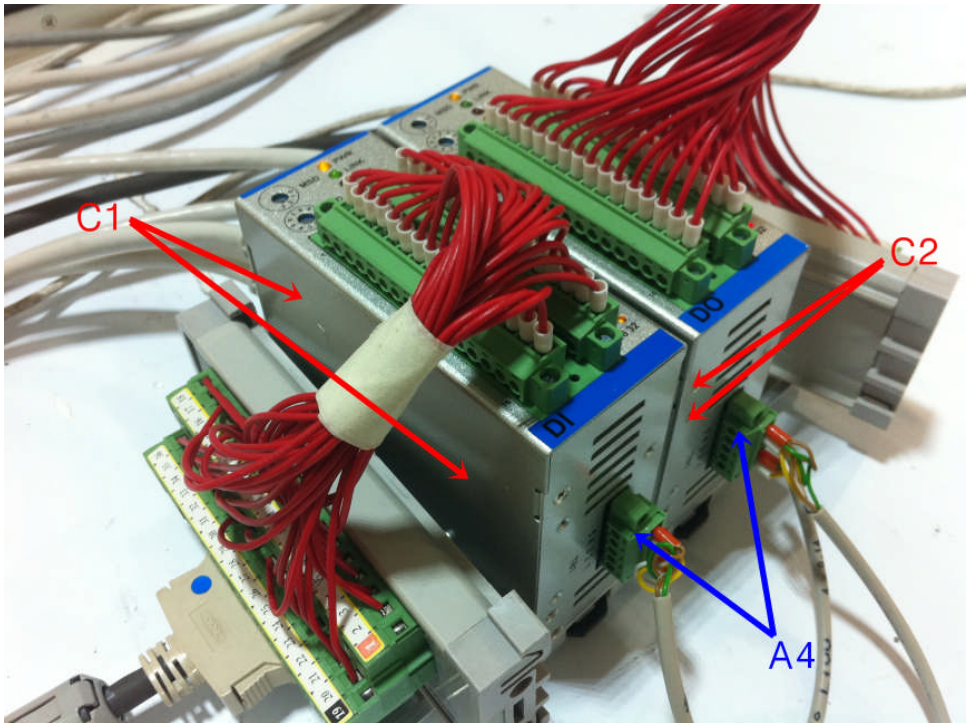
Air discharge →

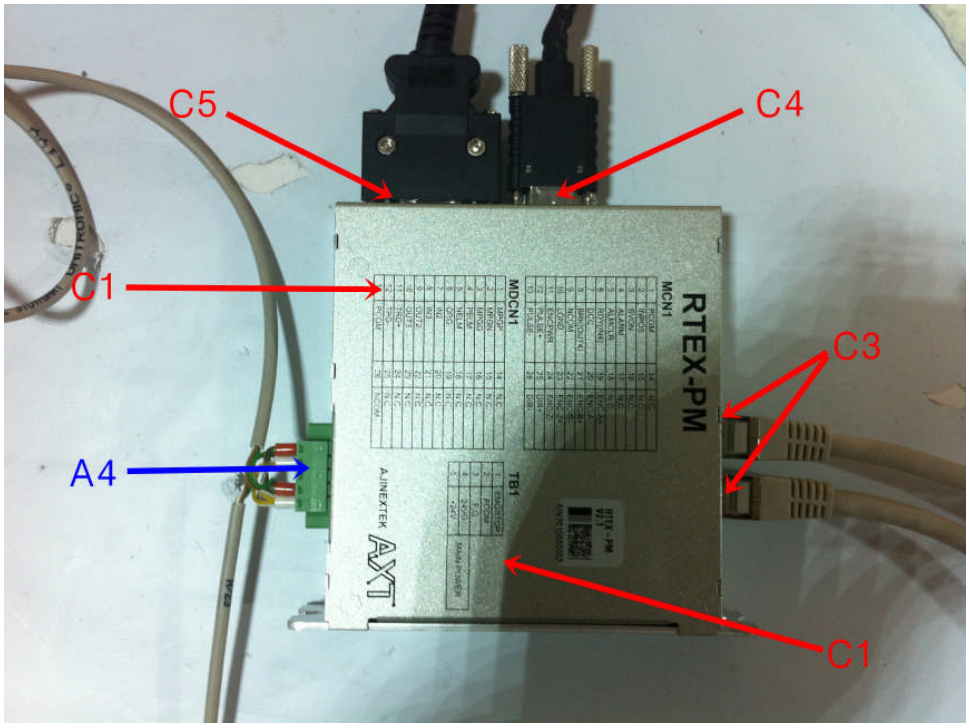
Contact discharge →





This test report shall not be reproduced, except in full, without the written approval.





This test report shall not be reproduced, except in full, without the written approval.

Contact discharge

Location(EUT)		Applied level (±)	Result (Criterion)
C1	Enclosure	± 2 kV, ± 4 kV	A
C2	Screw	± 2 kV, ± 4 kV	A
C3	LAN(RJ-45) Port	± 2 kV, ± 4 kV	A
C4	MCN1 Port	± 2 kV, ± 4 kV	A
C5	MDCN1 Port	± 2 kV, ± 4 kV	A
	HCP (All 4 sides)	± 2 kV, ± 4 kV	A
	VCP (All 4 sides)	± 2 kV, ± 4 kV	A

Air discharge

Location(EUT)		Applied level (±)	Result (Criterion)
A1	LED	± 2 kV, ± 4 kV, ± 8 kV	A
A2	DCN1 Port	± 2 kV, ± 4 kV, ± 8 kV	A
A3	Control Switch	± 2 kV, ± 4 kV, ± 8 kV	A
A4	TB1 Port	± 2 kV, ± 4 kV, ± 8 kV	A

This test report shall not be reproduced, except in full, without the written approval.

6.4 Radio Frequency Electromagnetic Fields

Test specification	EN 61000-4-3:2006, Criteria : A				
Tested frequency	80 ㎒ ~ 1 ㎓ log 1 % step				
Test level & Modulation	3 V/m, 80 % Amplitude Modulation (1 ㎑)				
Distance	3 m from EUT to tip of antenna				
Dwell time	3 s				
Step size	log 1 % step				
Testing Voltage	AC main, DC 24 V				
Date	2011. 07. 22				
Temperature(°C)	25 °C	Humidity (% R.H.)	40 % R.H.	Pressure (㎪)	100.2 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.4.1 Measurement procedure

The test was performed at 3 m full anechoic chamber.

For floor standing equipment, the EUT was standing on the floor.

For tabletop equipment, the EUT was located on a wooden table 0.8 m above the floor.

The EUT was tested all sides, horizontal and vertical polarization.

6.4.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Power meter	PM2002	302852	AR	12.04.11	☒
Power sensor	PH2000	303224	AR	12.04.11	☒
Power sensor	PH2000	303222	AR	12.04.11	☒
Directional coupler	DC6180	303976	AR	12.04.11	☒
Signal generator	E4421B	GB40052295	AGILENT	11.10.25	☒
Amplifier	150W1000M2	303843	AR	12.04.11	☒
Broadband Ant.	LPDA-0803	130269	ETS	-	☒
Field monitor	SI-300	-	TDK	-	☒
Antenna master	-	-	ETS	-	☒

This test report shall not be reproduced, except in full, without the written approval.

6.4.3 Photographs of test setup



6.4.4 Measurement result

Location(EUT)	Antenna polarization	Result (Criterion)
Front side	Horizontal	A
	Vertical	A
Rear side	Horizontal	A
	Vertical	A
Left side	Horizontal	A
	Vertical	A
Right side	Horizontal	A
	Vertical	A

This test report shall not be reproduced, except in full, without the written approval.

6.5 Electric Fast Transient/BURST

Test specification	EN 61000-4-4:2004, Criteria : B				
Coupling	AC main, DC 24 V Signal: Clamp				
Test level	AC main: ± 1 ㎸ Peak DC 24 V, Signal: ± 0.5 ㎸ Peak				
Repetition frequency	5 ㎑, Tr/Th = 5 / 50 ㎱				
Coupling time	60 s				
Testing Voltage	AC main, DC 24 ㅿ				
Date	2011. 07. 25				
Temperature(°C)	26 °C	Humidity (% R.H.)	39 % R.H.	Pressure (㎪)	100.1 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.5.1 Measurement procedure

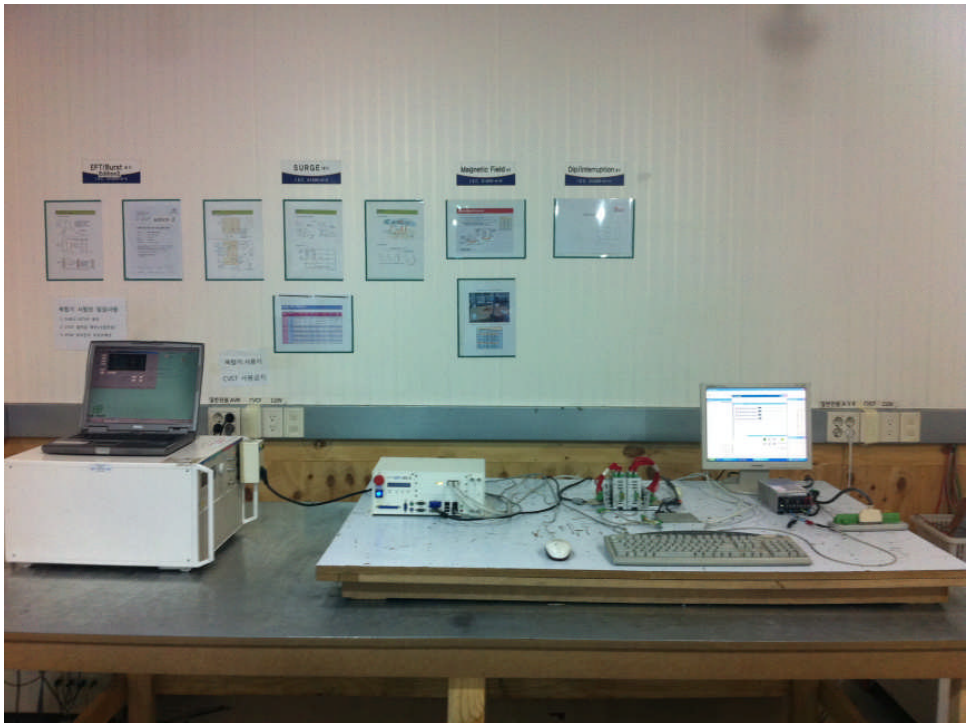
A ground reference plane was located on the floor.
EFT generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.1 ㎜ wooden table.
For tabletop equipment, EUT was placed on a 0.1 ㎜ above the ground reference plane.
Test generator and coupling/decoupling network was placed on, and bounded to, the ground reference plane. When using the coupling clamp, the minimum distance between the coupling plates and all other conductive surfaces, except the ground reference plane beneath the coupling clamp, Shall be 0.5 ㎜.

6.5.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	12.02.09	☒
Capacitive Coupling Clamp	-	-	EM TEST	-	☒

This test report shall not be reproduced, except in full, without the written approval.

6.5.3 Photographs of test setup





This test report shall not be reproduced, except in full, without the written approval.

6.5.4 Measurement result

* AC main

EFT coupling point	(+)	(-)	Result (Criterion)
L+N+PE	+ 1 kV	- 1 kV	A

* DC 24 V

EFT coupling point	(+)	(-)	Result (Criterion)
L+N	+ 0.5 kV	- 0.5 kV	A

* Signal (AC main, DC 24 V)

EFT coupling point	(+)	(-)	Result (Criterion)
RJ-45	+ 0.5 kV	- 0.5 kV	A

This test report shall not be reproduced, except in full, without the written approval.

6.6 Surge

Test specification	EN 61000-4-5:2006, Criteria : B				
Coupling	AC main				
Test level	AC main Differential mode: ± 0.5 kV, ± 1 kV Common mode: ± 0.5 kV, ± 1 kV, ± 2 kV				
Surge pulse shape	Tr/Th = 1.2 / 50 μs				
Coupling Impedance	AC main Differential mode: 18 μF Common mode: 10 Ω + 9 μF				
Angles	0 °, 90 °, 180 °, 270 °				
Number of surge / Coupling time	5 / 1 min				
Testing Voltage	AC main				
Date	2011. 07. 25				
Temperature(°C)	26 °C	Humidity (% R.H)	39 % R.H	Pressure(kPa)	100.1 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.6.1 Measurement procedure

A ground reference plane was located on the floor. SURGE generator was connected to reference ground plane via low impedance connection. For floor standing equipment & table top equipment, EUT was placed on a wooden table.

6.6.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	12.02.09	☒
CDN	CNV 508 S1	0302-01	EM TEST	-	☐

This test report shall not be reproduced, except in full, without the written approval.

6.6.3 Photographs of test setup



6.6.4 Measurement result

* AC main

Coupling point	(+)	(-)	Result (Criterion)
L-N	+ 0.5 kV, + 1 kV	- 0.5, - 1 kV	A
L-PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A
N-PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A

This test report shall not be reproduced, except in full, without the written approval.

6.7 Conducted Immunity

Test specification	EN 61000-4-6:2007, Criteria : A				
Tested frequency	0.15 ㎒ ~ 80 ㎒ log 1 % step				
Test level & Modulation	1, 3, 10 Ⅴ, 80 %Amplitude Modulation (1 ㎑)				
Coupling method	AC main: M3 (CDN) DC 24 Ⅴ: M2(CDN) Signal : Clamp				
Testing Voltage	AC main, DC 24 Ⅴ				
Date	2011. 07. 22				
Temperature(°C)	26 °C	Humidity (% R.H)	38 % R.H	Pressure(㎪)	100.2 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.7.1 Measurement procedure

A ground reference plane was located on the floor.

The test was performed on a ground reference plane on a 0.1 m wooden table. This test were performed using CDN for mains, clamp for signal and injection probe. The frequency range was swept from 0.15 ㎒ to 80 ㎒. This frequency range was Modulated with 1 ㎑ sine wave at 80 %.

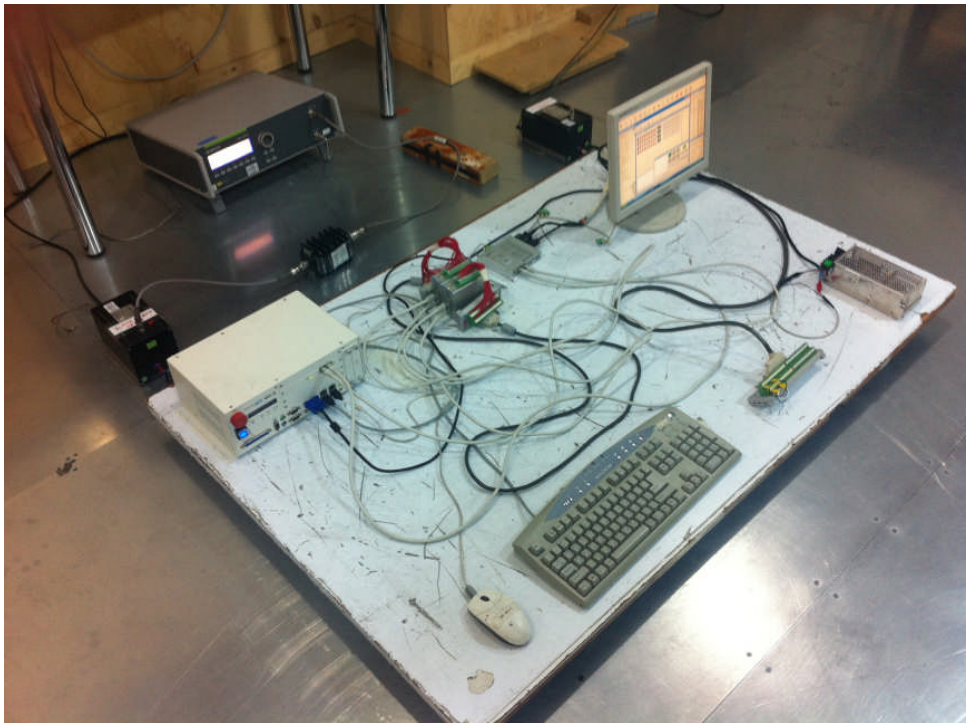
The signal generators provided the modulated frequency at a 1 % step size. The power and all network cable, I/O cables longer than 3 m length were tested.

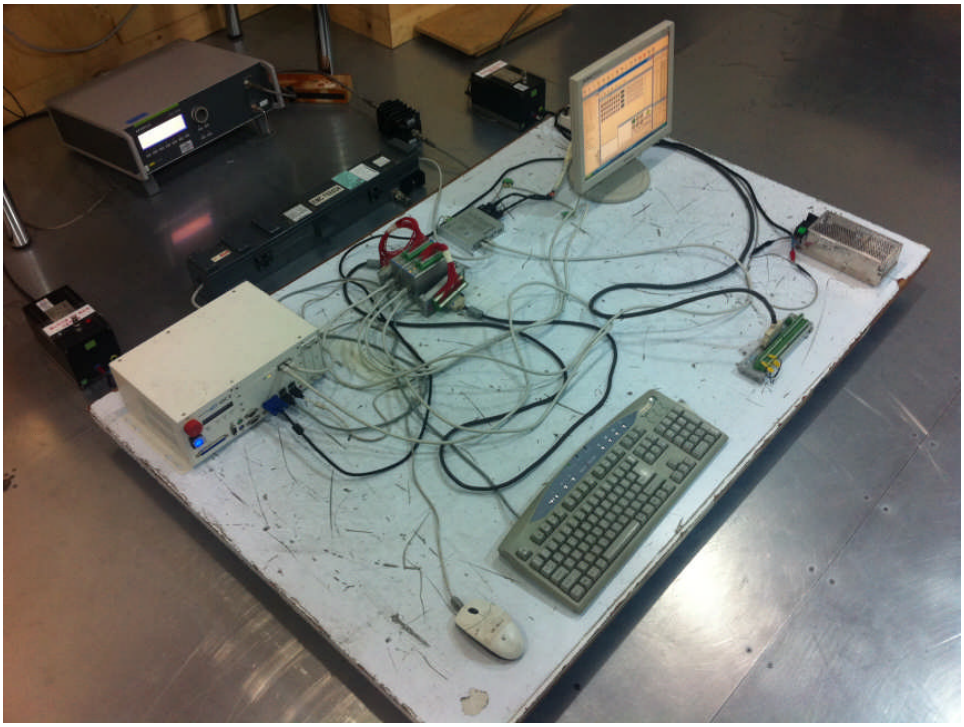
6.7.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
CS generator	CWS 500 C S1	V0635101750	EM TEST	11.10.13	☒
CDN	CDN M2/M3	0906-12	EM TEST	11.10.13	☒
CDN	CDN M3-32A	0506-29	EM TEST	12.02.11	☒
Attenuator	73-6-34	MU918	MCE/WEINSCHEL	11.10.13	☒
EM Clamp	KEMZ 801	17643	Schaffner	12.04.25	☒
Current probe	MD720	W1345167/M6/ 0068	Schaffner	-	☐

This test report shall not be reproduced, except in full, without the written approval.

6.7.3 Photographs of test setup





This test report shall not be reproduced, except in full, without the written approval.

6.7.4 Measurement result

* AC main

Coupling point	Coupling method	Result (Criterion)
Power	CDN (M3)	A

* DC 24 V

Coupling point	Coupling method	Result (Criterion)
Power	CDN (M2)	A

* Signal (AC main, DC 24 Ⅴ)

Coupling point	Coupling method	Result
RJ-45	Clamp	A

This test report shall not be reproduced, except in full, without the written approval.

6.8 Dips and Interruptions

Test specification	EN 61000-4-11:2004 , Criterion: B or C				
Number of dips	3 T				
Duration	10 s				
Phase	Zero crossing (0 °, 180 °)				
Testing Voltage	AC main				
Date	2011. 07. 25				
Temperature (°C)	26 °C	Humidity (% R.H)	39 % R.H	Pressure (㎪)	100.1 ㎪
Remarks	Complied				

6.8.1 Measurement procedure

The dips/interruption test is only applicable to AC mains.

The dips/interruptions were applied at zero crossing.

6.8.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	12.02.09	☒

This test report shall not be reproduced, except in full, without the written approval.

6.8.3 Photographs of test setup



6.10.4 Measurement result

* 230 V, 50 Hz

Test Level (%UT)	Dip/Int. (%UT)	Duration /Period	Angle (°)	Count number	Result
0 %	100 %	0.5 Period	0 / 180	3T	A
70 %	30 %	25 Period	0	3T	A
0 %	100 %	250 Period	0	3T	C

Comment:

- A: There was no change of operation status during above testing. (0.5 Period, 25 Period)
- C: The power of EUT is off during the interruption test.
  After the test, When EUT is rebooting, the EUT is also getting back to normal operation.
  (250 Period)

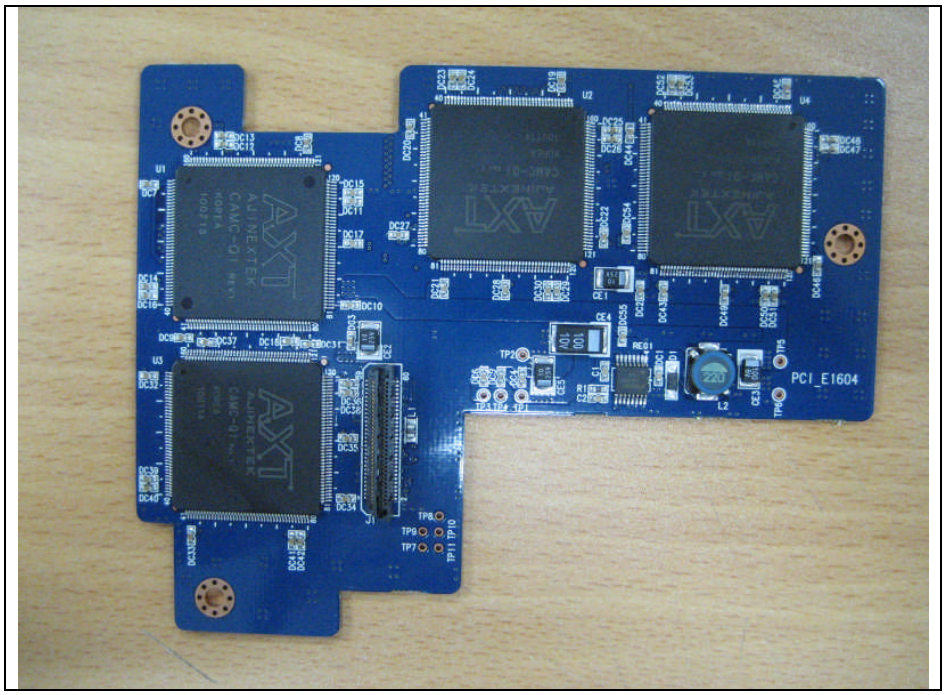
This test report shall not be reproduced, except in full, without the written approval.

7. E.U.T. photographs

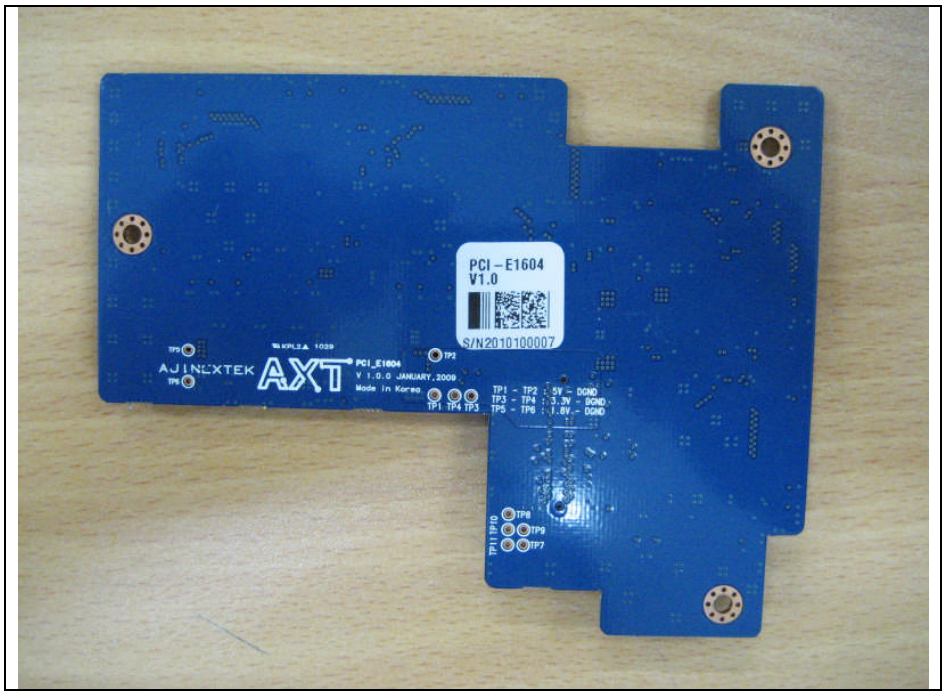
-EUT(Main)

PCI-E1604 Board

Front View



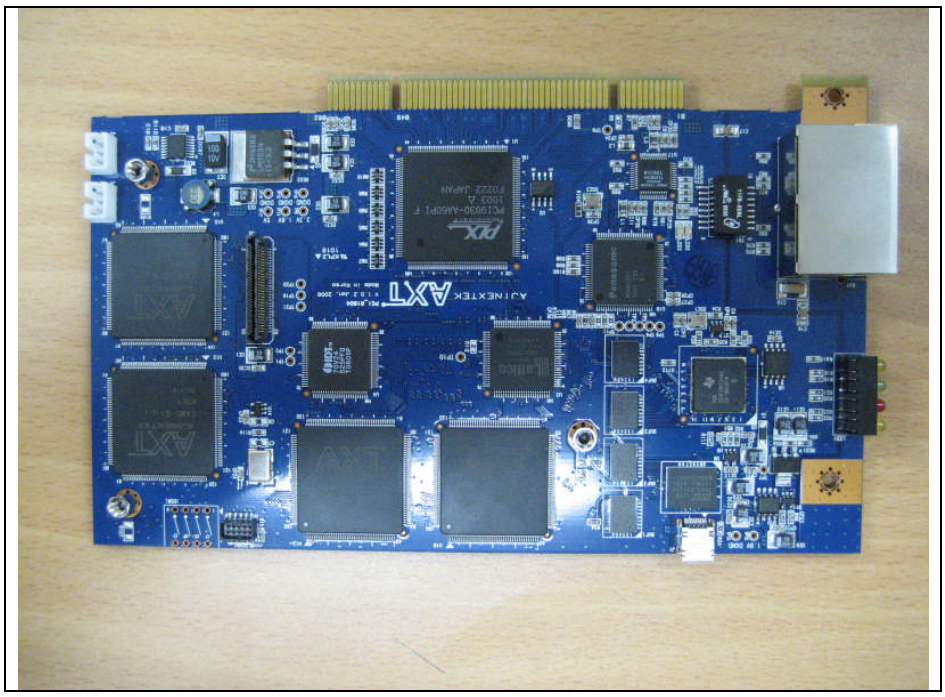
Rear View



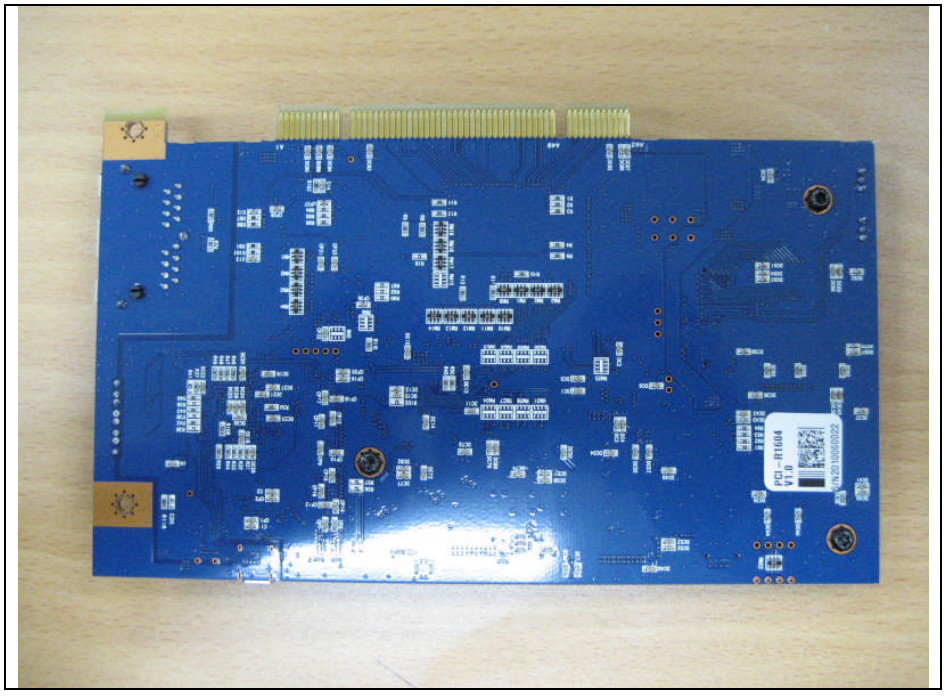
This test report shall not be reproduced, except in full, without the written approval.

PCI-R1604 Board

Front View



Rear View



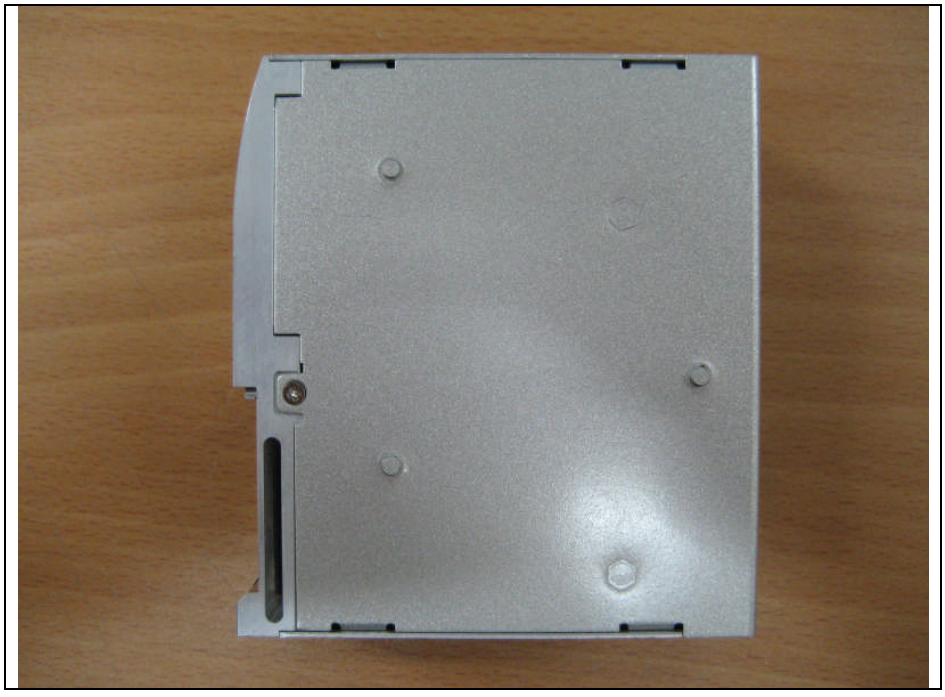
This test report shall not be reproduced, except in full, without the written approval.

- RTEX-DI32(EUT)

Top View

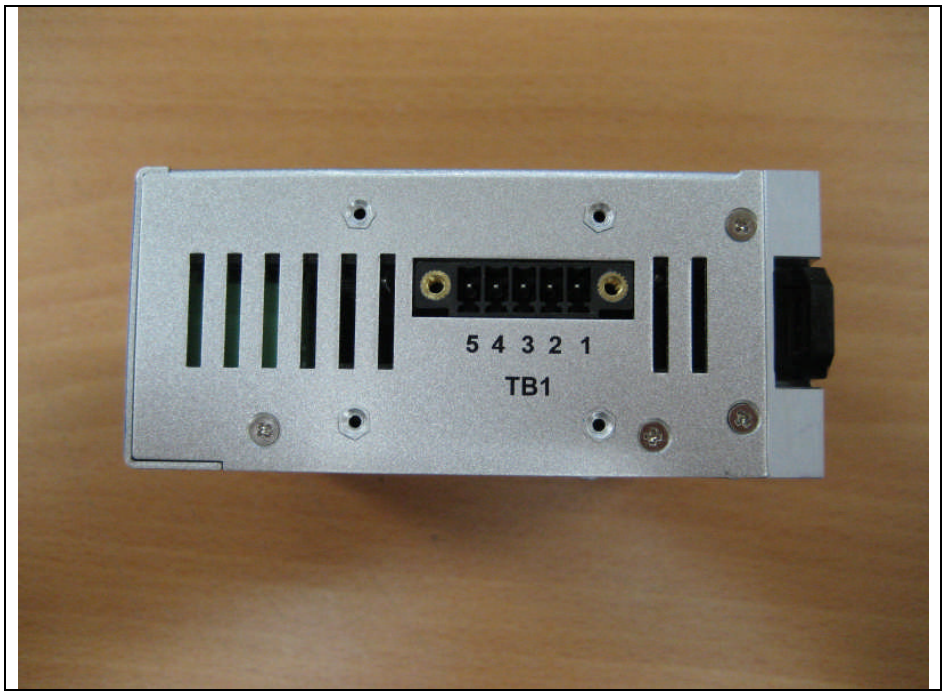


Bottom View

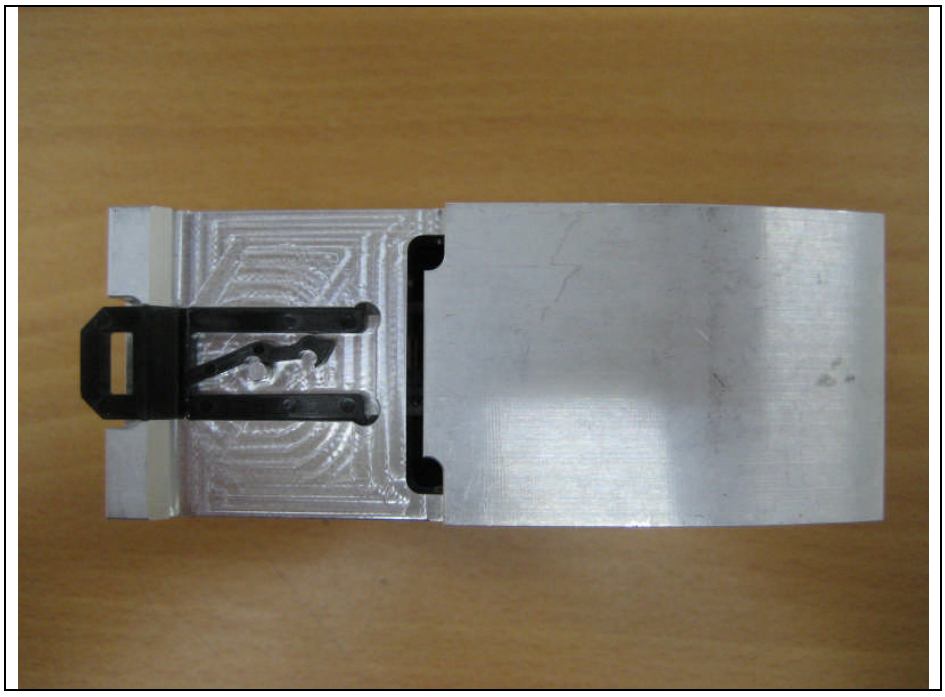


This test report shall not be reproduced, except in full, without the written approval.

Left View

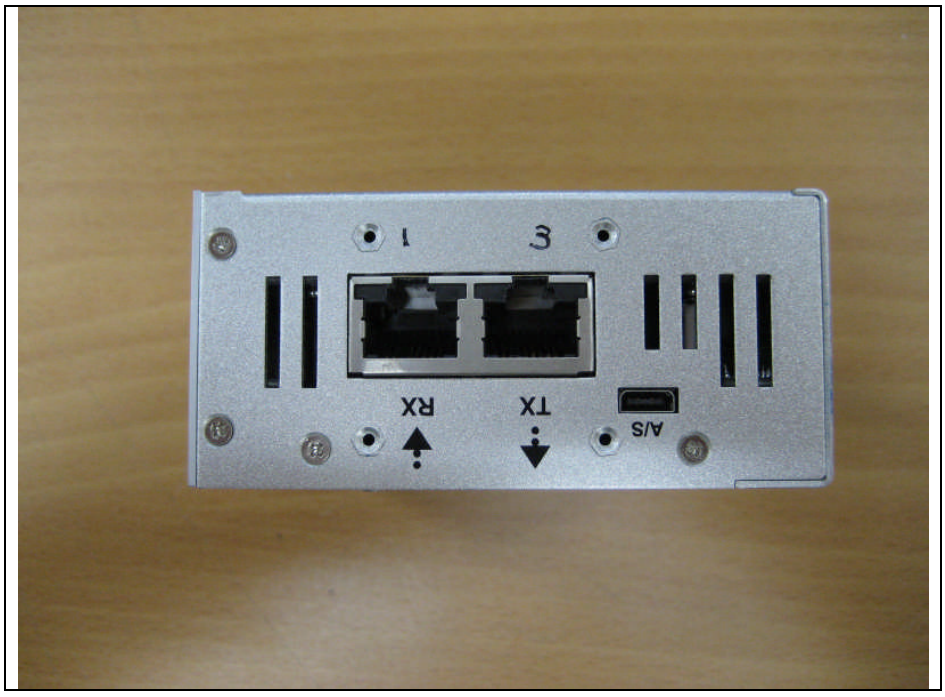


Right View

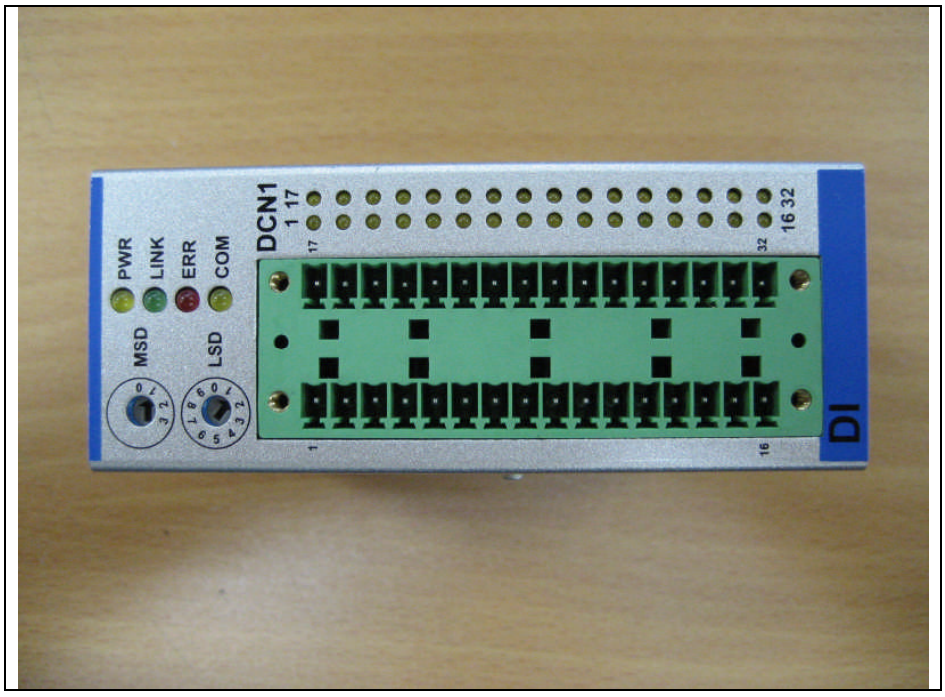


This test report shall not be reproduced, except in full, without the written approval.

Top View

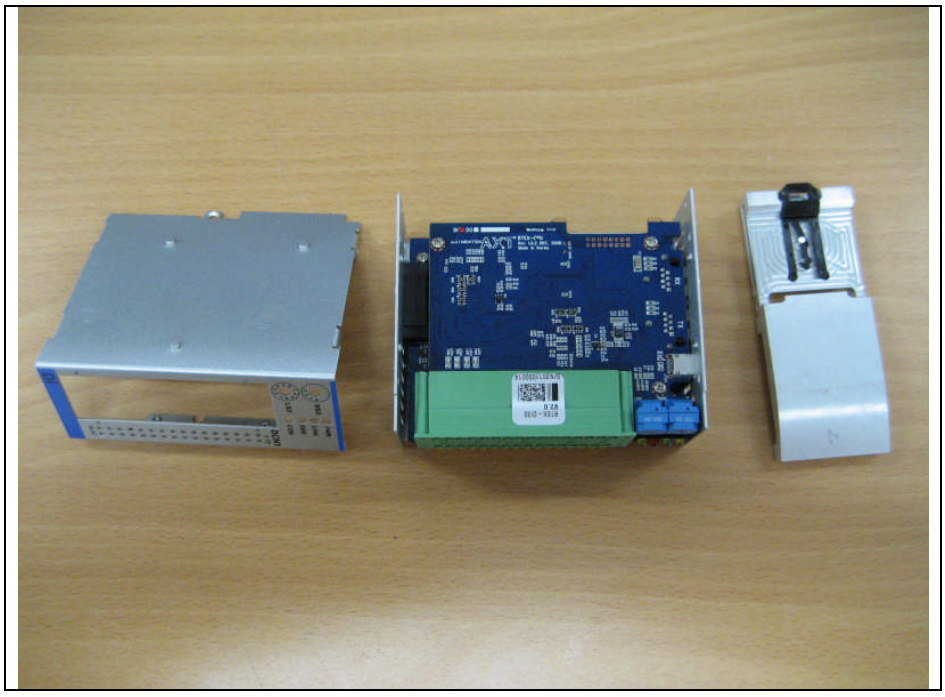


Bottom View



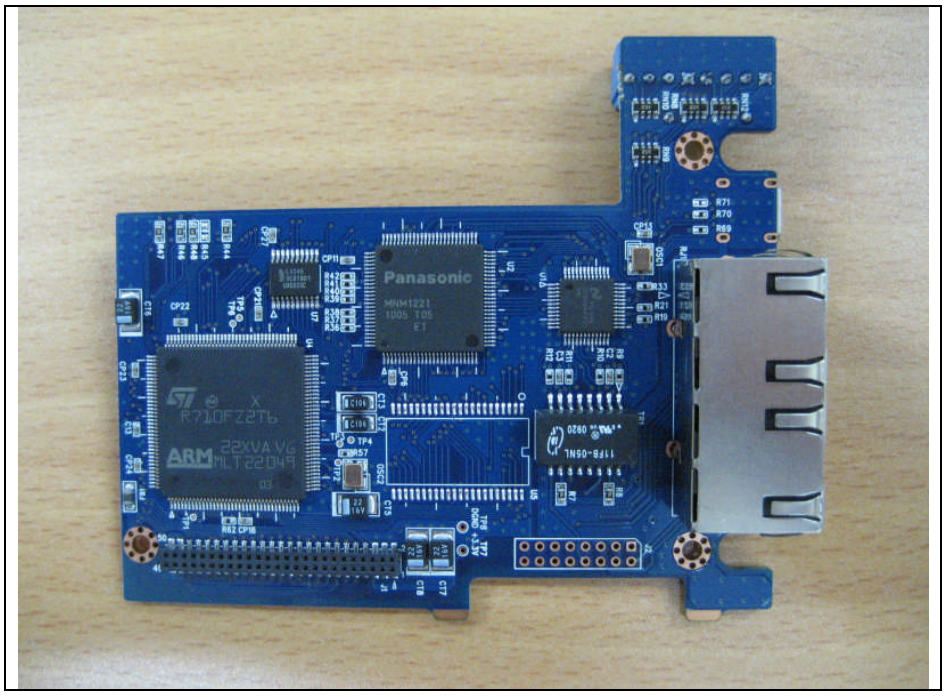
This test report shall not be reproduced, except in full, without the written approval.

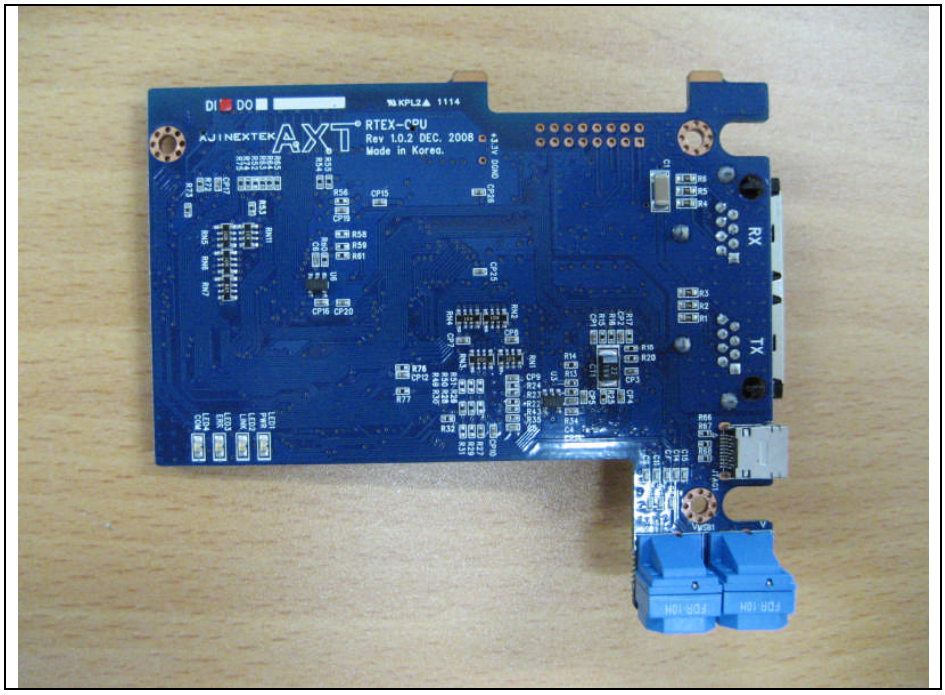
Inside



This test report shall not be reproduced, except in full, without the written approval.

RTEX-CPU Board





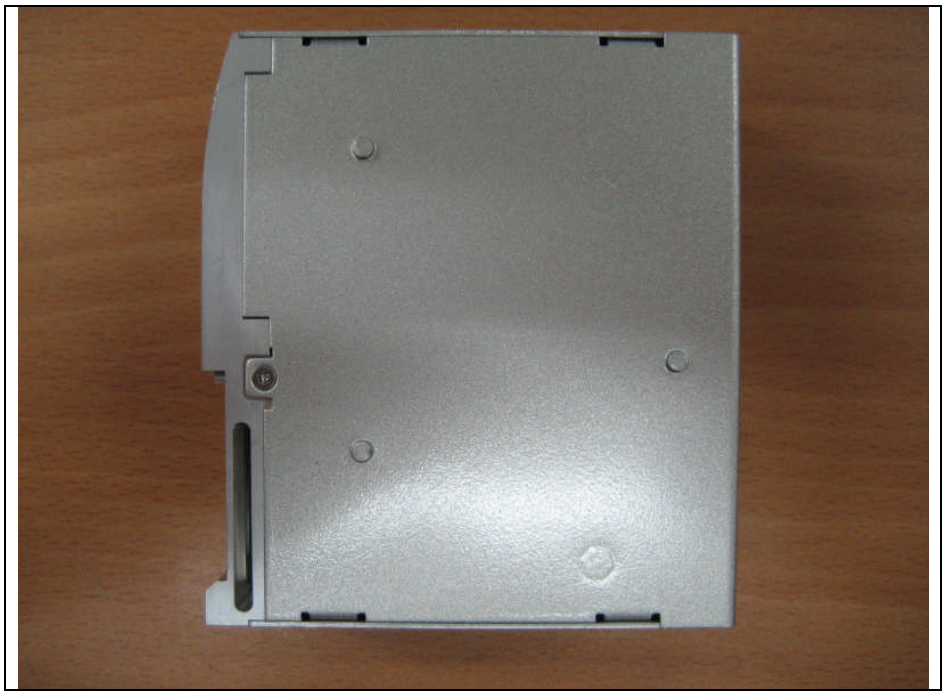
This test report shall not be reproduced, except in full, without the written approval.

- RTEX-DO32T(EUT)

Front View

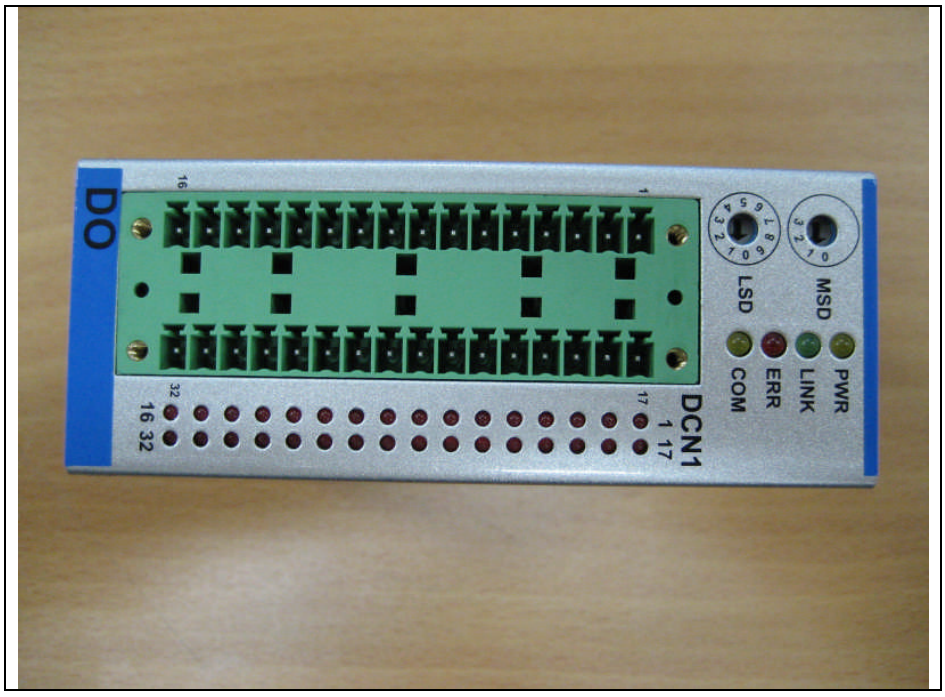


Rear View

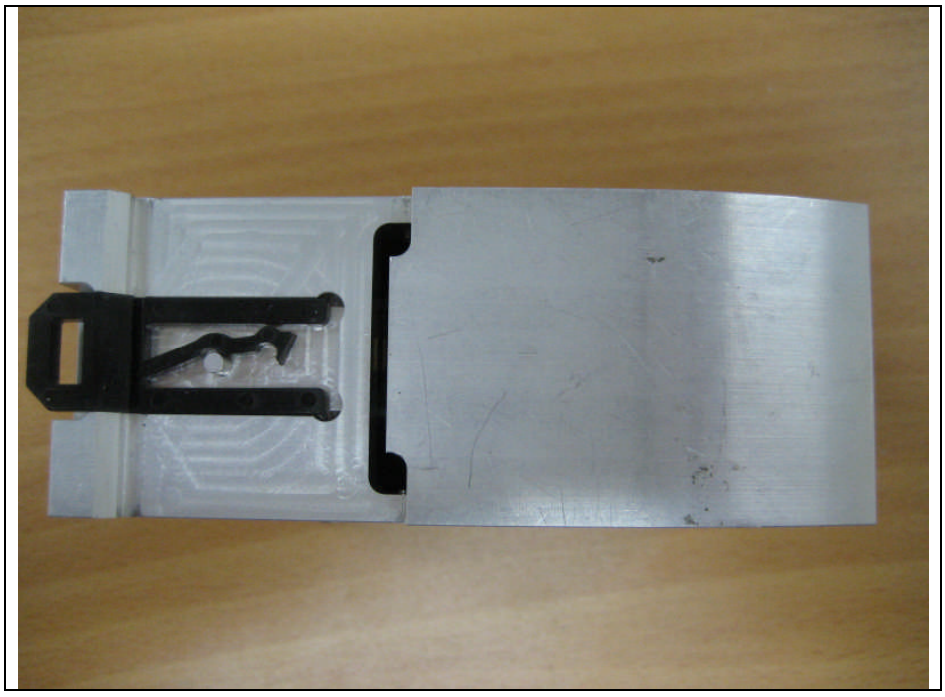


This test report shall not be reproduced, except in full, without the written approval.

Left View

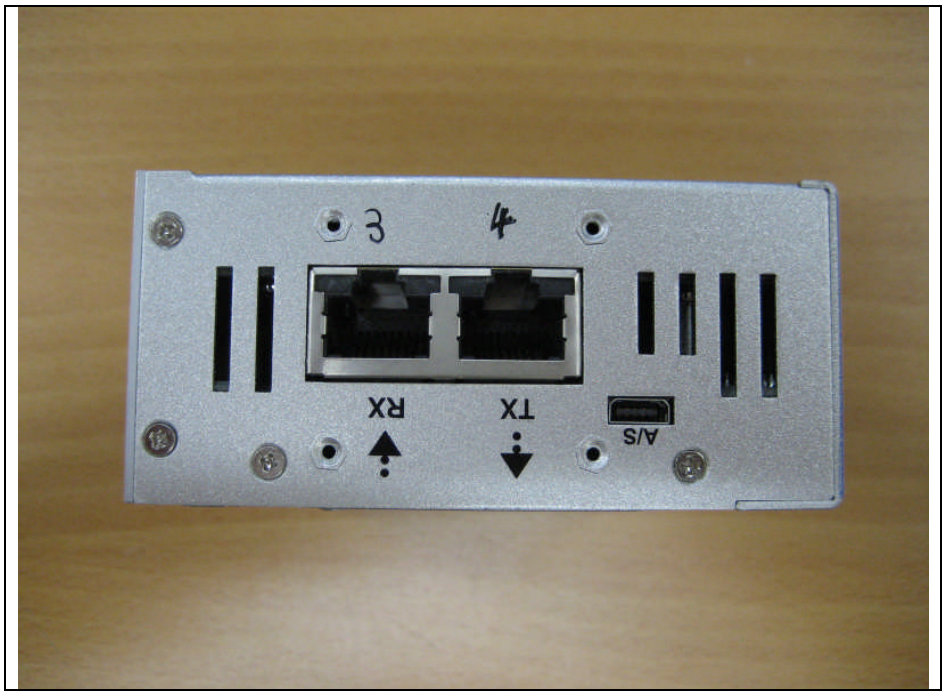


Right View

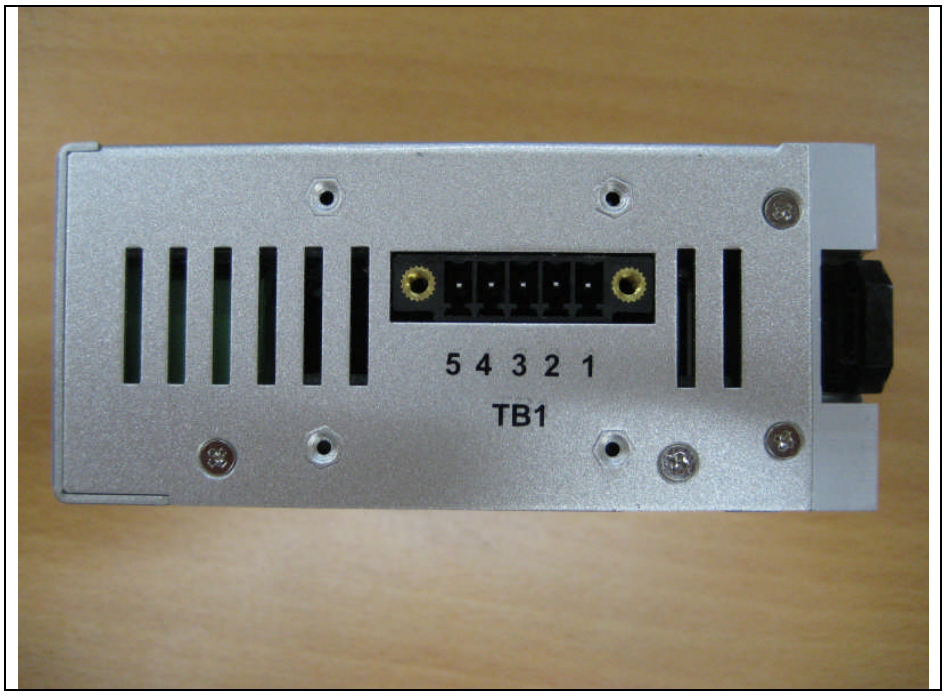


This test report shall not be reproduced, except in full, without the written approval.

Top View

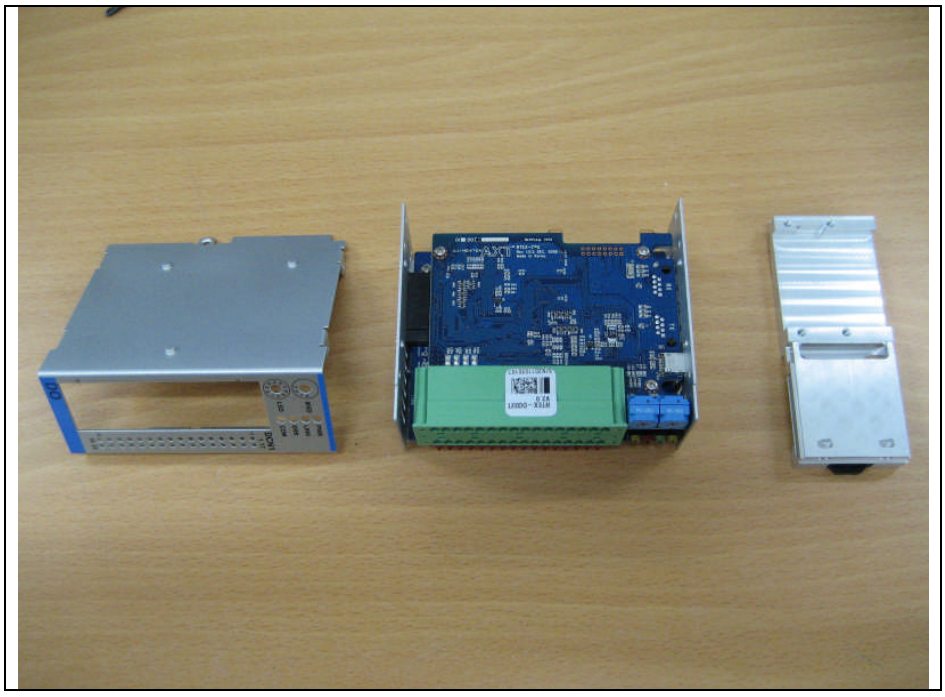


Bottom View



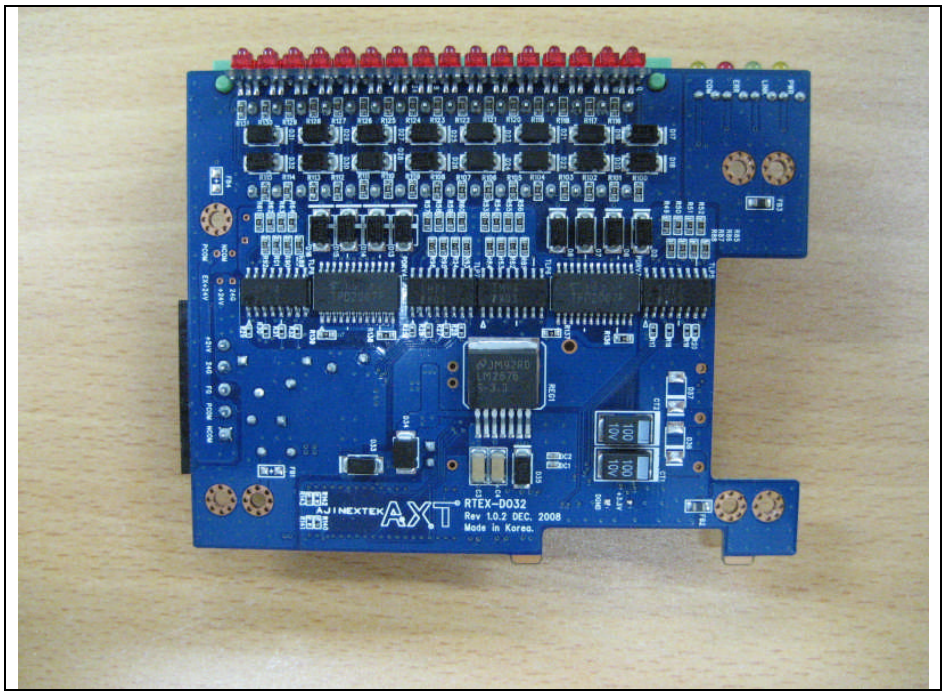
This test report shall not be reproduced, except in full, without the written approval.

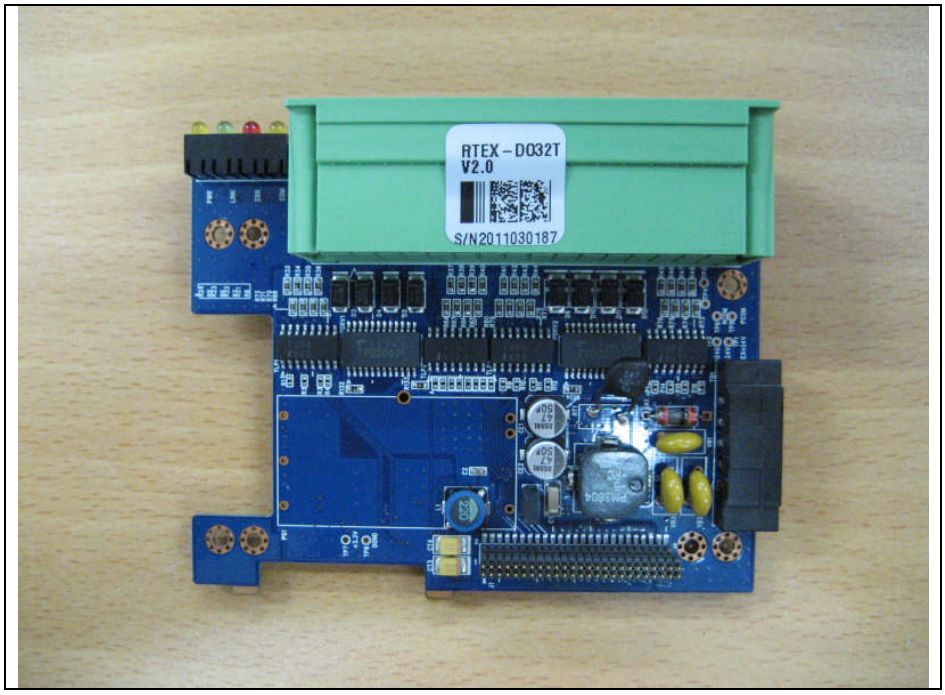
Inside



This test report shall not be reproduced, except in full, without the written approval.

RTEX-DO32 Board

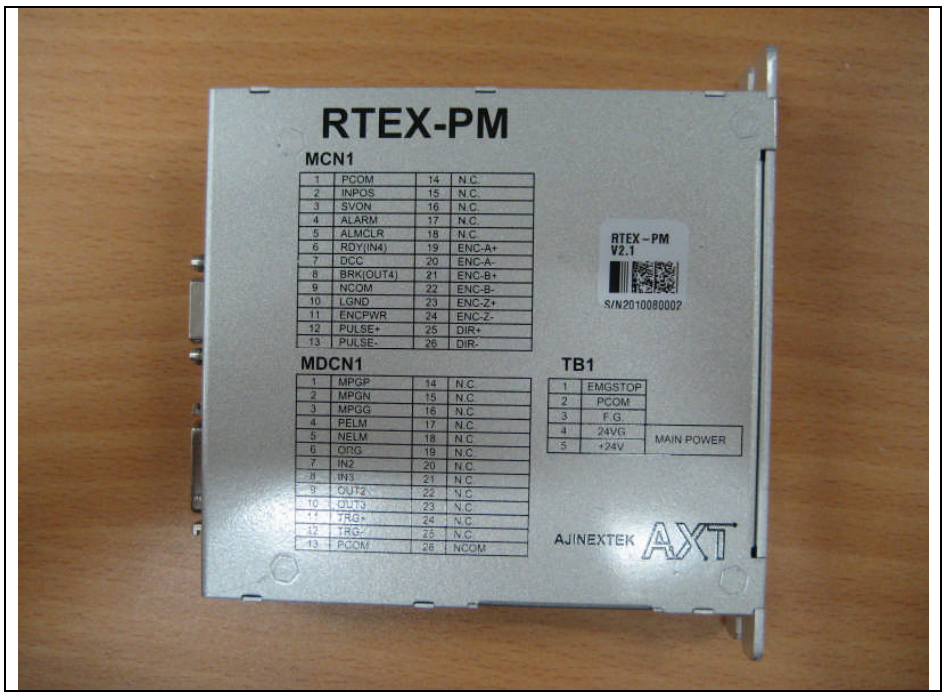




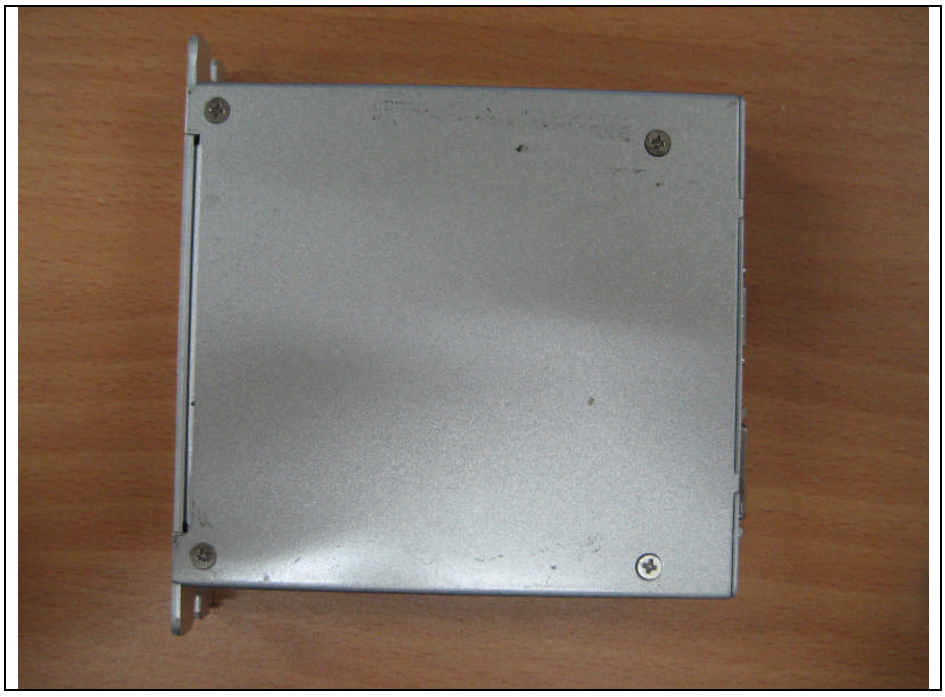
This test report shall not be reproduced, except in full, without the written approval.

- RTEX-PM(EUT)

Front View

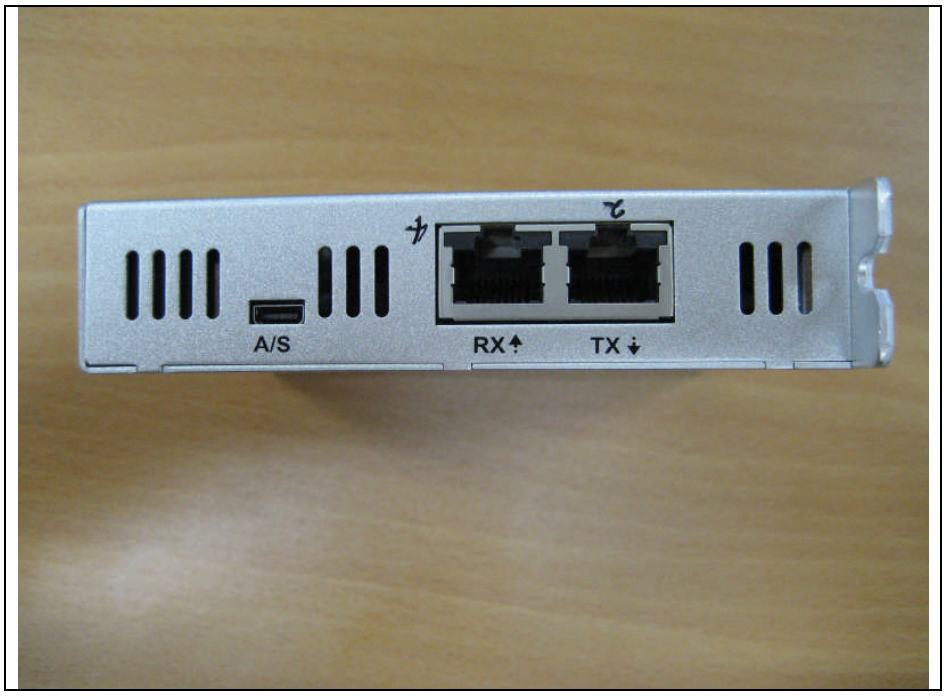


Rear View



This test report shall not be reproduced, except in full, without the written approval.

Left View

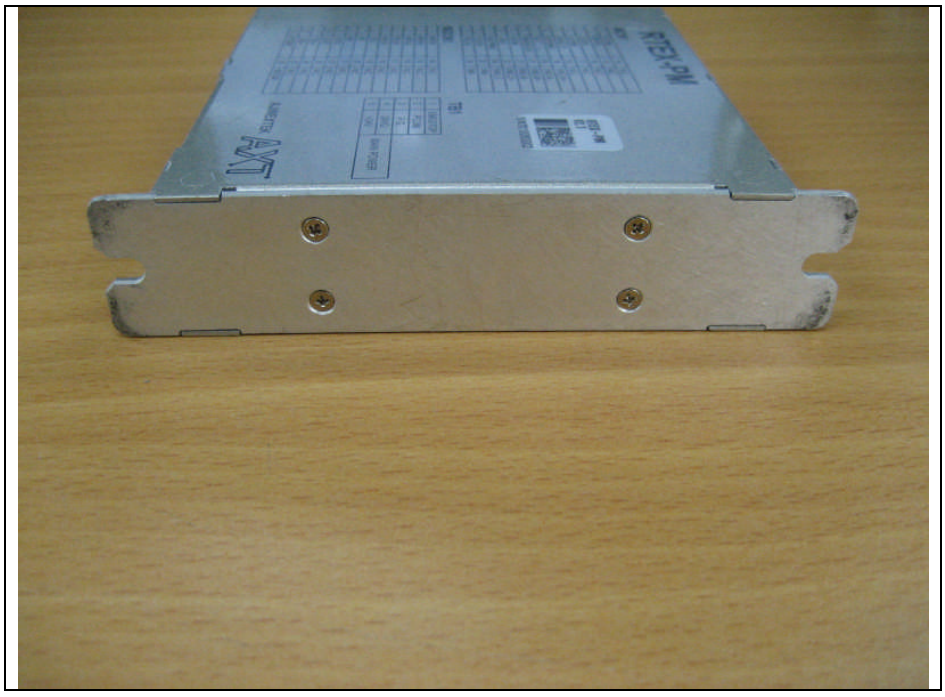


Right View

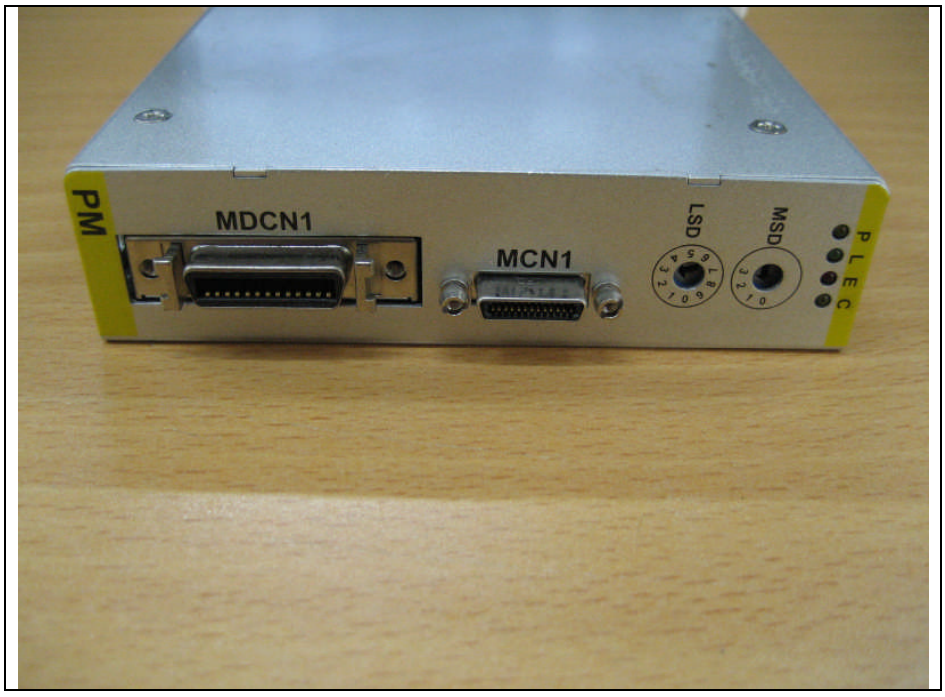


This test report shall not be reproduced, except in full, without the written approval.

Top View

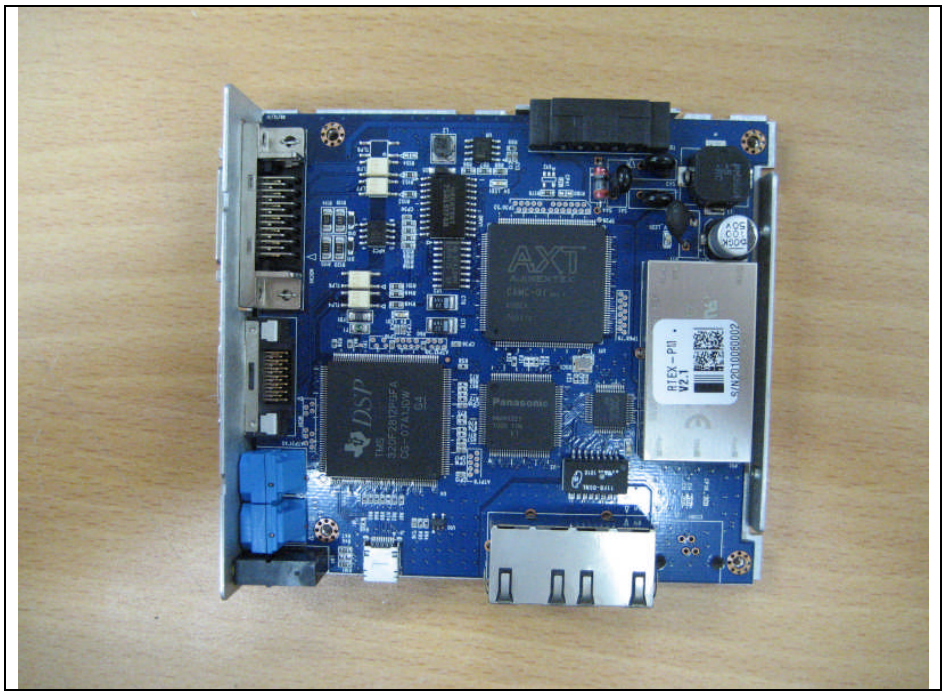


Bottom View



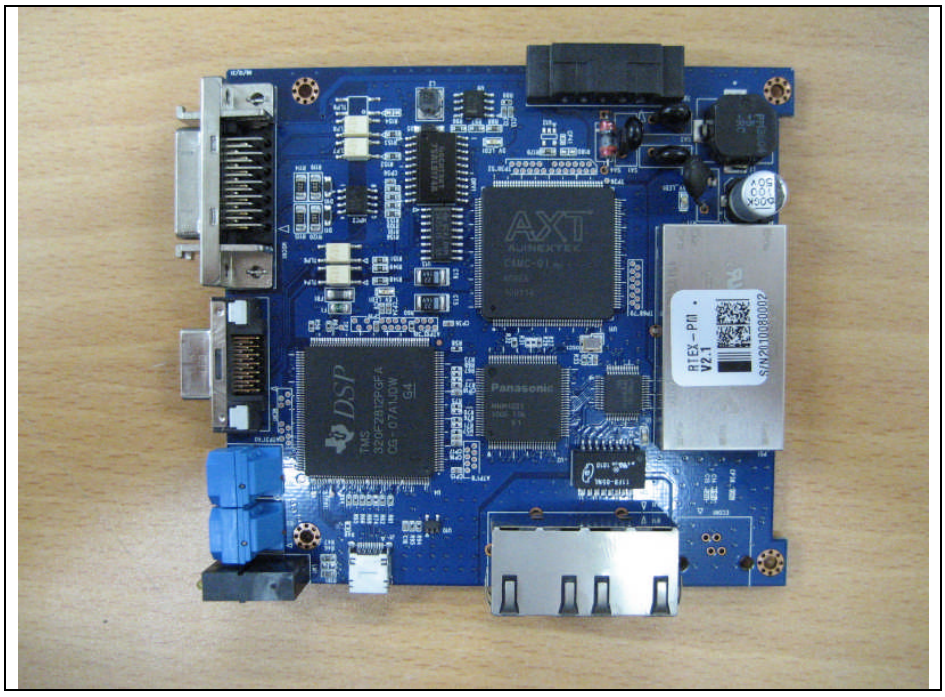
This test report shall not be reproduced, except in full, without the written approval.

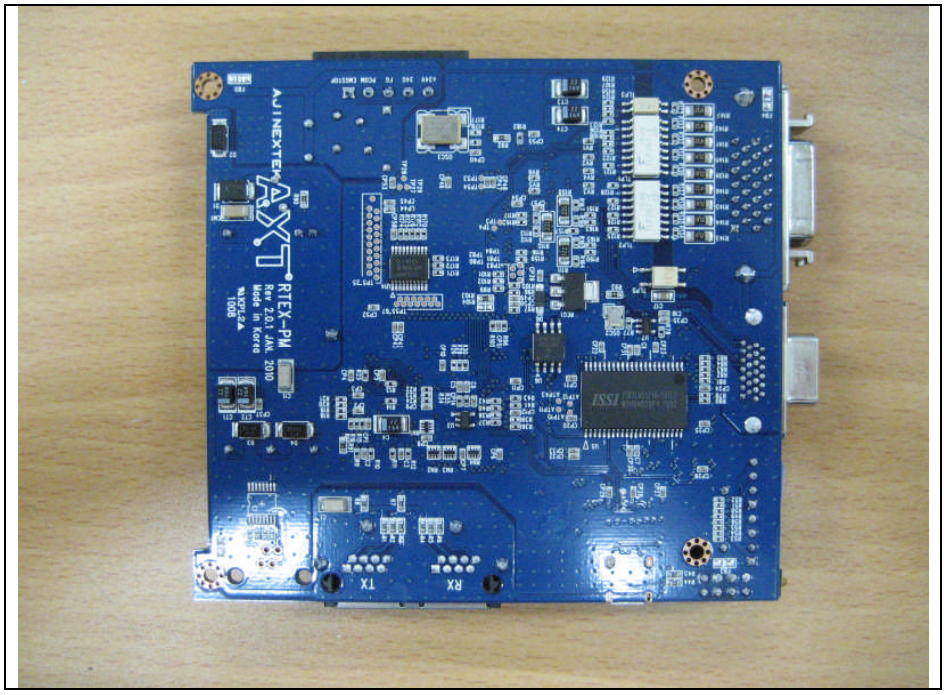
Inside



This test report shall not be reproduced, except in full, without the written approval.

RTEX-PM Board





This test report shall not be reproduced, except in full, without the written approval.